Kinematic analysis of rigid-link Spatial Mechanisms

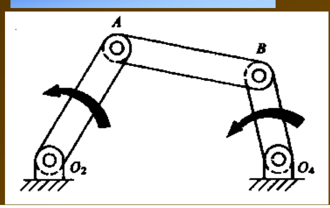
Basics of Mechanism

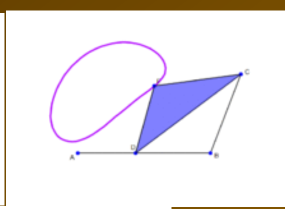
Classification of Mechanism

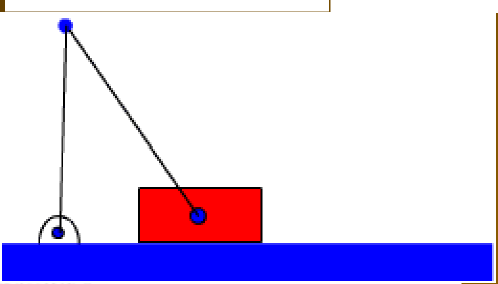
Planar Mechanism

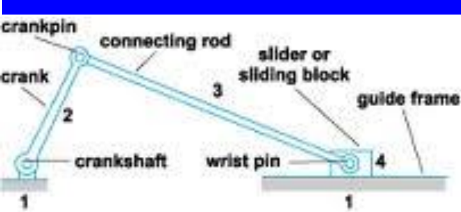
All the points of a mechanism move in parallel planes

Closed Chain







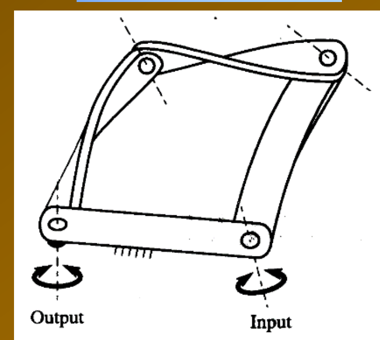


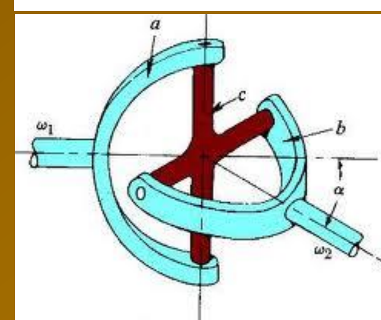
Open Chain

Spatial Mechanism

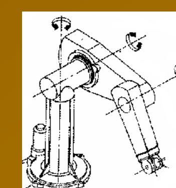
All the points of a mechanism do not move in parallel planes

Closed Chain





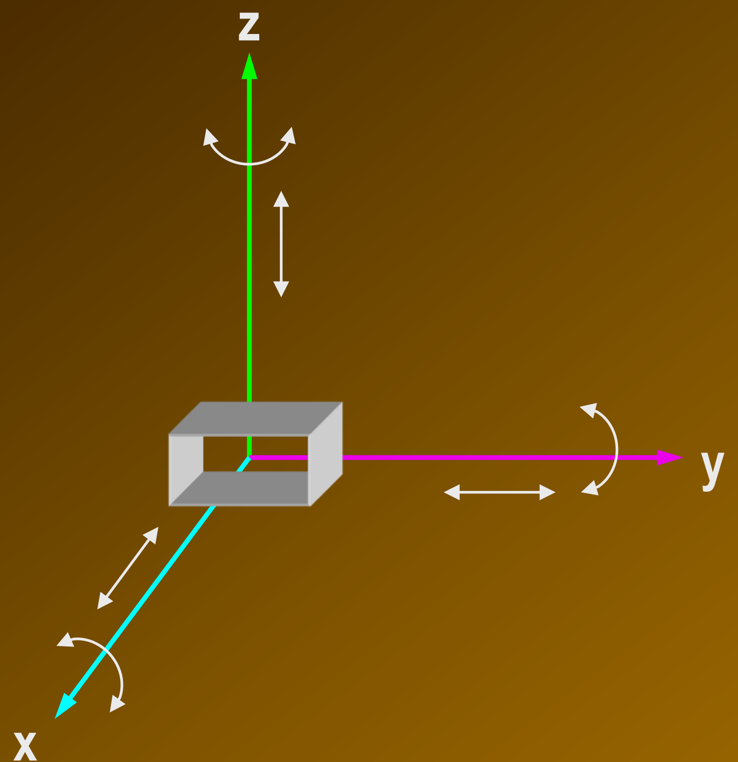
Open Chain

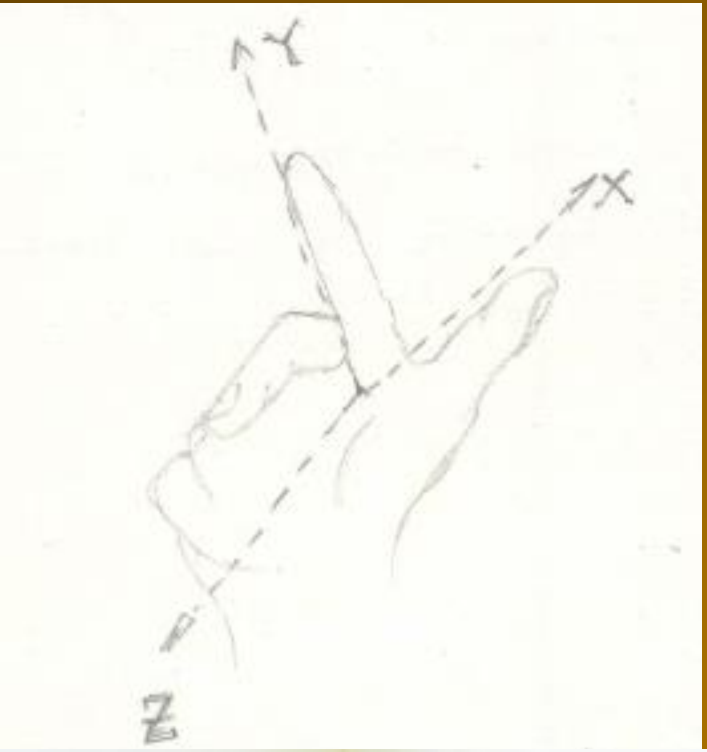


Spatial Geometry

Fundamentals of Spatial Geometry

Description of a POINT in space



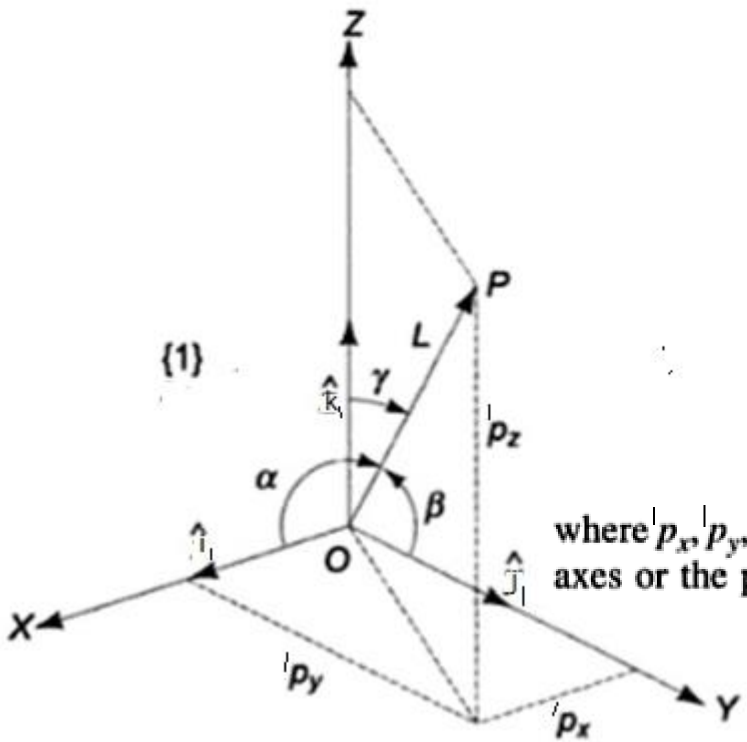


RH Cartesian coordinate system or
RH frame

Fundamentals of Spatial Geometry

Description of a POINT in space

Representation in Cartesian Coordinate system



$$ {}^1P = {}^1p_x\,\hat{i} + {}^1p_y\,\hat{j} + {}^1p_z\,\hat{k}, $$

$$ {}^1P = \begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} = [\,{}^1p_x \quad {}^1p_y \quad {}^1p_z\,]^T $$

where ${}^1p_x, {}^1p_y, {}^1p_z$ are the components of the vector $\overrightarrow{OP}$ along the three coordinate axes or the projections of the vector $\overrightarrow{OP}$ on the axes X, Y, Z, respectively.

$$ \cos\alpha = \frac{{}^1p_x}{L}, \cos\beta = \frac{{}^1p_y}{L}, \cos\gamma = \frac{{}^1p_z}{L} $$

$$ L = |\vec{P}| = |\overrightarrow{OP}| = \sqrt{({}^1p_x)^2 + ({}^1p_y)^2 + ({}^1p_z)^2} $$

where α, β, and γ are, respectively, the right handed angles measured from the coordinate axes to the vector $\overrightarrow{OP}$, which has a length L.

Description of a POINT in space

Representation in Homogeneous Coordinate

In homogeneous co-ordinate, a point 'p' in space (3-D) is denoted as

$$\vec{P} = \begin{bmatrix} x' \\ y' \\ z' \\ \sigma \end{bmatrix} \qquad {}^{'}P = \begin{bmatrix} (\sigma' p_x) \\ (\sigma' p_y) \\ (\sigma' p_z) \\ \sigma \end{bmatrix} \Rightarrow 4 \times 1 \text{ matrix} \quad ((n+1) \dim.)$$

- the fourth component 'σ' is a non-zero positive scaling factor.

The physical co-ordinates in Cartesian representation are obtained by dividing each component in homogeneous representation by the scaling factor.

If the value of the scaling factor (σ) is set to '1', the components of homogeneous & Cartesian representation are identical.

Description of a POINT in space

Representation in Homogeneous Coordinate

In homogeneous co-ordinate, a point 'p' in space (3-D) is denoted as

$$\vec{P} = \begin{bmatrix} x' \\ y' \\ z' \\ \sigma \end{bmatrix} \qquad {}^1P = \begin{bmatrix} (\sigma \cdot {}^1p_x) \\ (\sigma \cdot {}^1p_y) \\ (\sigma \cdot {}^1p_z) \\ \sigma \end{bmatrix} \Rightarrow 4\times 1 \text{ matrix} \quad ((n+1) \text{ dim.})$$

- the fourth component 'σ' is a non-zero positive scaling factor.

For example: physical position vector

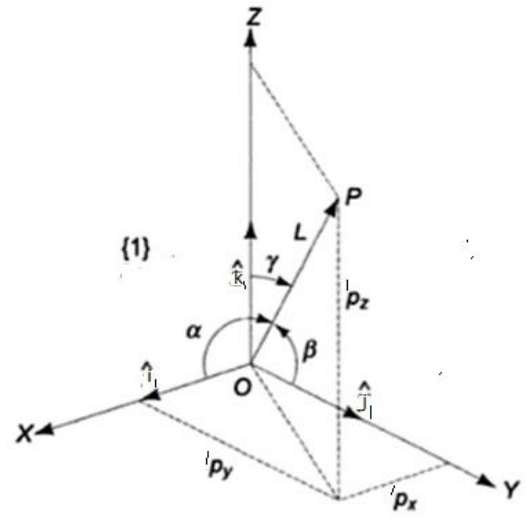
$$\vec{P} = \cancel{8}\ 3\hat{i} - 2\hat{j} + 4\hat{k} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$$

In homogeneous co-ordinate

$\vec{P}$ equivalent to $\vec{P_1} = \begin{bmatrix} 3 \\ -2 \\ 4 \\ 1 \end{bmatrix}$ with $\sigma = 1$ (Same as physical co-ordinate)

$= \begin{bmatrix} 6 \\ -4 \\ 8 \\ 2 \end{bmatrix}$ with $\sigma = 2$

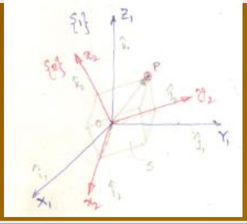
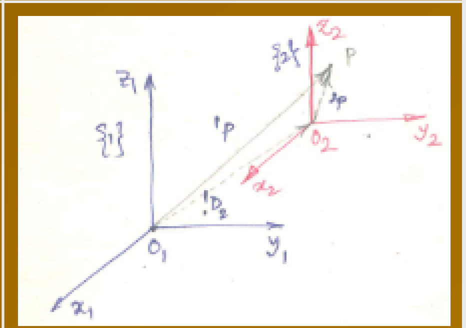
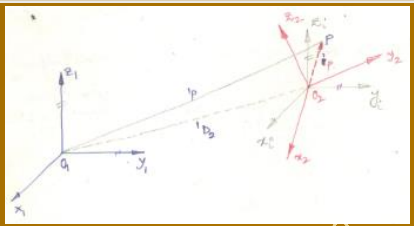
$= \begin{bmatrix} 1.5 \\ -1 \\ 2 \\ 0.5 \end{bmatrix}$ with $\sigma = 0.5$



Mapping

Mapping refers to changing the description of a POINT (or a vector) from one coordinate system (i.e. frame) to another coordinate system (i.e. frame).

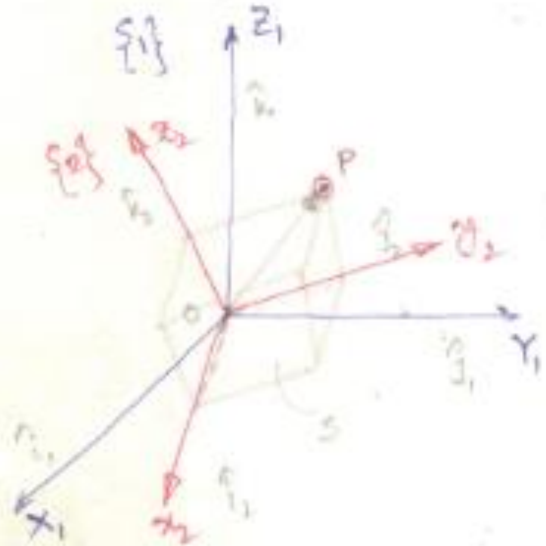
Second coordinate frame has THREE possibilities in relation to the first coordinate frame

(i)	Second coordinate frame is rotated w.r.to the first but origin of both the frames is same	**Mapping between Rotated frame**	
(ii)	Second coordinate frame is moved away from first, the axes of both frames remains parallel. This is a translation of the origin of the second frame from the first frame.	**Mapping between Translated frame**	
(iii)	Second coordinate frame is rotated as well as translated	**Mapping between Rotated & Translated frame**	

Mapping

Mapping between Rotated Frame

Second coordinate frame is rotated w.r.to the first but origin of both the frames is same



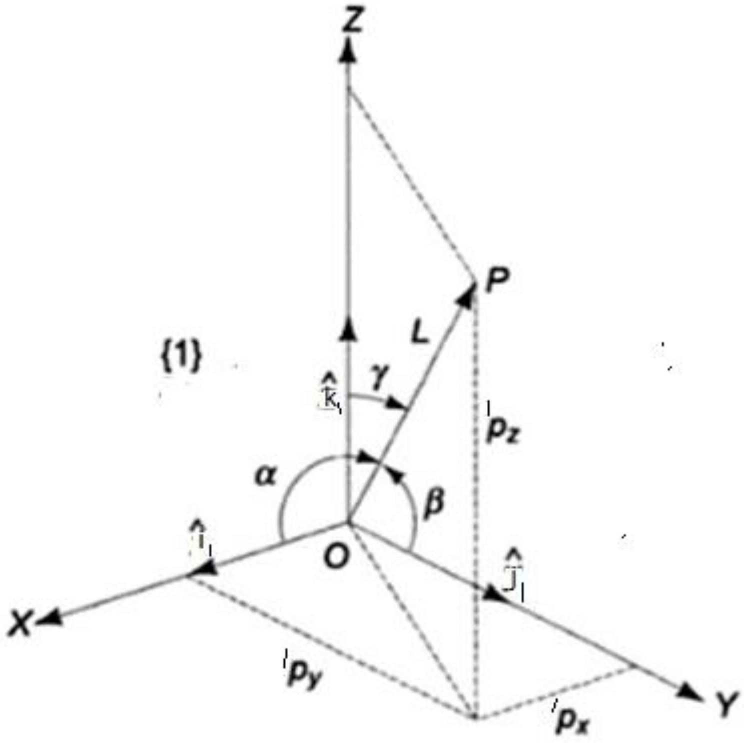
Consider a rigid body 'S' to which a Co-ordinate frame {2} with axes {x_2, y_2, z_2} is attached.
Frame {2} is a body attached frame.
We wish to relate the Co-ordinates of a point 'p' on rigid body S in frame {2} to the Co-ordinates of 'p' in a fixed reference frame {1}.

Both the frame {1} & {2} have a common origin.

A point 'p' in space can be described by the two frames {1} & {2} can be expressed as vectors ^1p and ^2p

w.r.to frame {1} → $^1P = {}^1p_x \cdot \hat{i}_1 + {}^1p_y \cdot \hat{j}_1 + {}^1p_z \cdot \hat{k}_1$ or $^1p = \begin{bmatrix} ^1p_x \\ ^1p_y \\ ^1p_z \end{bmatrix}$

w.r.to frame {2} → $^2P = {}^2p_x \cdot \hat{i}_2 + {}^2p_y \cdot \hat{j}_2 + {}^2p_z \cdot \hat{k}_2$ or $^2p = \begin{bmatrix} ^2p_x \\ ^2p_y \\ ^2p_z \end{bmatrix}$

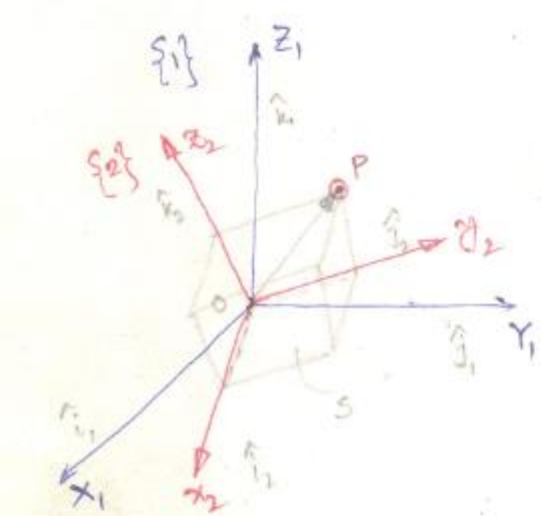


$$\cos\alpha = \frac{{}^1p_x}{L},\ \cos\beta = \frac{{}^1p_y}{L},\ \cos\gamma = \frac{{}^1p_z}{L}$$

$$L = |\vec{P}| = |\overrightarrow{OP}| = \sqrt{({}^1p_x)^2 + ({}^1p_y)^2 + ({}^1p_z)^2}$$

Mapping between Rotated Frame

Second coordinate frame is rotated w.r.to the first but origin of both the frames is same



Since ^1p & ^2p are representations of the same vector $\vec{OP}$, the relationship betn. the components of $\vec{OP}$ in the two co-ordinate frames can be obtained.

$$^1p = \vec{OP} \quad \text{and} \quad ^2p = \vec{OP}$$

st Problem

Aim: The description of point 'P' in frame {2} is known & its description in frame {1} is to be found.

$$\left[{}^1p_x = |{}^1p||\hat{i}_1|\cos\alpha \quad (\Leftarrow {}^1P\cdot\hat{i}_1) \atop = |{}^1p|\cos\alpha. \right]$$

$$^1p_x = {}^1P\cdot\hat{i}_1 = |{}^1p||\hat{i}_1|\cos\alpha_1$$

$$^1p_y = |{}^1p||\hat{j}_1|\cos\beta_1$$

$$^1p_z = |{}^1p||\hat{k}_1|\cos\gamma_1$$

$$\cos\alpha = \frac{^1p_x}{L},\ \cos\beta = \frac{^1p_y}{L},\ \cos\gamma = \frac{^1p_z}{L}$$

$$L = |\vec{P}| = |\vec{OP}| = \sqrt{(^1p_x)^2 + (^1p_y)^2 + (^1p_z)^2}$$

$\overrightarrow{OP}\cdot\hat{i}$

$${}^1p_x = {}^1p \cdot \hat{i}_1 = {}^2p \cdot \hat{i}_1 = \left({}^2p_x \hat{i}_2 + {}^2p_y \cdot \hat{j}_2 + {}^2p_z \cdot \hat{k}_2\right) \cdot \hat{i}_1 = {}^2p_x \cdot \hat{i}_2 \cdot \hat{i}_1 + {}^2p_y \hat{j}_2 \cdot \hat{i}_1 + {}^2p_z \hat{k}_2 \cdot \hat{i}_1$$

dot product

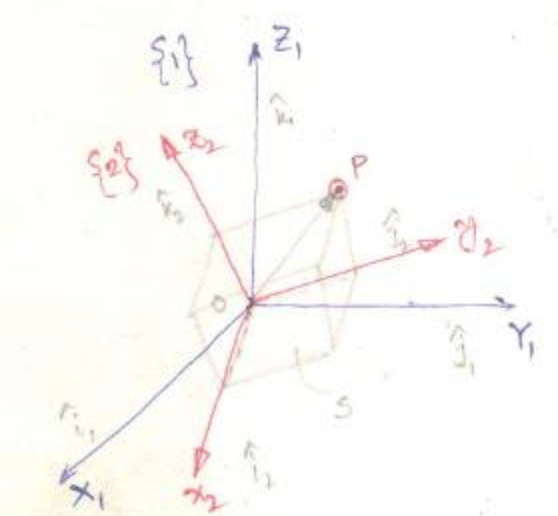
$${}^1p_y = {}^1p \cdot \hat{j}_1 = {}^2p \cdot \hat{j}_1 = \left({}^2p_x \cdot \hat{i}_2 + {}^2p_y \cdot \hat{j}_2 + {}^2p_z \cdot \hat{k}_2\right) \cdot \hat{j}_1 = {}^2p_x \cdot \hat{i}_2 \cdot \hat{j}_1 + {}^2p_y \cdot \hat{j}_2 \cdot \hat{j}_1 + {}^2p_z \cdot \hat{k}_2 \cdot \hat{j}_1$$

$${}^1p_z = {}^1p \cdot \hat{k}_1 = {}^2p \cdot \hat{k}_1 = \left({}^2p_x \cdot \hat{i}_2 + {}^2p_y \cdot \hat{j}_2 + {}^2p_z \cdot \hat{k}_2\right) \hat{k}_1 = {}^2p_x \cdot \hat{i}_2 \cdot \hat{k}_1 + {}^2p_y \cdot \hat{j}_2 \cdot \hat{k}_1 + {}^2p_z \hat{k}_2 \cdot \hat{k}_1$$

Mapping

Mapping between Rotated Frame

Second coordinate frame is rotated w.r.to the first but origin of both the frames is same



First Problem

Aim: The description of point 'P' in frame {2} is known & its description in frame {1} is to be found.

$$\left[\begin{aligned} {}^1p_x &= |{}^1p|\,|\hat{i}_1|\cos\alpha \quad (\Leftarrow {}^1p.\hat{i}_1) \\ &= |{}^1p|\cos\alpha. \end{aligned}\right]$$

dot product

$${}^1p_x = {}^1p.\hat{i}_1 = {}^2p.\hat{i}_1 = ({}^2p_x.\hat{i}_2 + {}^2p_y.\hat{j}_2 + {}^2p_z.\hat{k}_2).\hat{i}_1 = {}^2p_x.\hat{i}_2.\hat{i}_1 + {}^2p_y.\hat{j}_2.\hat{i}_1 + {}^2p_z.\hat{k}_2.\hat{i}_1$$

$${}^1p_y = {}^1p.\hat{j}_1 = {}^2p.\hat{j}_1 = ({}^2p_x.\hat{i}_2 + {}^2p_y.\hat{j}_2 + {}^2p_z.\hat{k}_2).\hat{j}_1 = {}^2p_x.\hat{i}_2.\hat{j}_1 + {}^2p_y.\hat{j}_2.\hat{j}_1 + {}^2p_z.\hat{k}_2.\hat{j}_1$$

$${}^1p_z = {}^1p.\hat{k}_1 = {}^2p.\hat{k}_1 = ({}^2p_x.\hat{i}_2 + {}^2p_y.\hat{j}_2 + {}^2p_z.\hat{k}_2)\hat{k}_1 = {}^2p_x.\hat{i}_2.\hat{k}_1 + {}^2p_y.\hat{j}_2.\hat{k}_1 + {}^2p_z.\hat{k}_2.\hat{k}_1$$

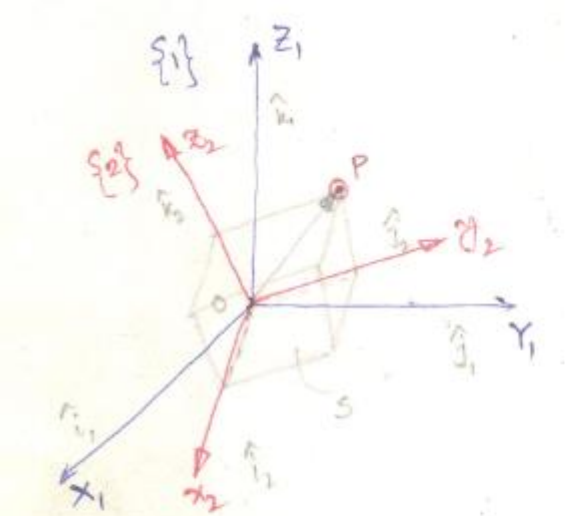
$$\begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} = \begin{array}{c} \\ X_1 \\ Y_1 \\ Z_1 \end{array} \begin{array}{c} \begin{array}{ccc} x_2 & y_2 & z_2 \end{array} \\ \begin{bmatrix} \hat{i}_2.\hat{i}_1 & \hat{j}_2.\hat{i}_1 & \hat{k}_2.\hat{i}_1 \\ \hat{i}_2.\hat{j}_1 & \hat{j}_2.\hat{j}_1 & \hat{k}_2.\hat{j}_1 \\ \hat{i}_2.\hat{k}_1 & \hat{j}_2.\hat{k}_1 & \hat{k}_2.\hat{k}_1 \end{bmatrix} \end{array} \begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix}$$

$${}^1P = {}^1_2R\,{}^2P \quad ---(2) \longrightarrow$$ Compressed vector-matrix notation/form.

Mapping

Mapping between Rotated Frame

Second coordinate frame is rotated w.r.to the first but origin of both the frames is same



$$
\begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} =
\begin{array}{c|ccc|}
 & x_2 & y_2 & z_2 \\
x_1 & \hat{i}_2 \cdot \hat{i}_1 & \hat{j}_2 \cdot \hat{i}_1 & \hat{k}_2 \cdot \hat{i}_1 \\
y_1 & \hat{i}_2 \cdot \hat{j}_1 & \hat{j}_2 \cdot \hat{j}_1 & \hat{k}_2 \cdot \hat{j}_1 \\
z_1 & \hat{i}_2 \cdot \hat{k}_1 & \hat{j}_2 \cdot \hat{k}_1 & \hat{k}_2 \cdot \hat{k}_1
\end{array}
\begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix}
$$

$$
{}^1R_2 = \begin{bmatrix} i_2 \cdot i_1 & j_2 \cdot i_1 & k_2 \cdot i_1 \\ i_2 \cdot j_1 & j_2 \cdot j_1 & k_2 \cdot j_1 \\ i_2 \cdot k_1 & j_2 \cdot k_1 & k_2 \cdot k_1 \end{bmatrix} \quad \text{------ (3)}
$$

= Rotation Matrix or Rotational Transformation matrix

- It contains only the dot products of unit vectors of the two frames & is independent of the point 'p'.
- ${}^1R_2$ can be used for transformation of the coordinates of any pt. 'P' in frame {2} to frame {1

Mapping

Mapping between Rotated Frame

$$\begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} = \begin{array}{c} \\ X_1 \\ Y_1 \\ Z_1 \end{array} \begin{array}{c} \begin{array}{ccc} x_2 & y_2 & z_2 \end{array} \\ \begin{bmatrix} \hat{i}_2 \cdot \hat{i}_1 & \hat{j}_2 \cdot \hat{i}_1 & \hat{k}_2 \cdot \hat{i}_1 \\ \hat{i}_2 \cdot \hat{j}_1 & \hat{j}_2 \cdot \hat{j}_1 & \hat{k}_2 \cdot \hat{j}_1 \\ \hat{i}_2 \cdot \hat{k}_1 & \hat{j}_2 \cdot \hat{k}_1 & \hat{k}_2 \cdot \hat{k}_1 \end{bmatrix} \end{array} \begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix}$$

Second Part: Aim: ${}^1P$ = known

$${}^2p_x = {}^2P \cdot i_2 = {}^1P \cdot i_2 = {}^1p_x\, i_1 \cdot i_2 + {}^1p_y\, j_1 \cdot i_2 + {}^1p_z\, k_1 \cdot i_2$$

$${}^2p_y = {}^2P \cdot j_2 = {}^1P \cdot j_2 = {}^1p_x\, i_1 \cdot j_2 + {}^1p_y\, j_1 \cdot j_2 + {}^1p_z \cdot k_1\, j_2$$

$${}^2p_z = {}^2P \cdot k_2 = {}^1P \cdot k_2 = {}^1p_x \cdot i_1 \cdot k_2 + {}^1p_y \cdot j_1 \cdot k_2 + {}^1p_z\, k_1 \cdot k_2$$

$$\begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix} = \begin{bmatrix} i_1 \cdot i_2 & j_1 \cdot i_2 & k_1 \cdot i_2 \\ i_1 \cdot j_2 & j_1 \cdot j_2 & k_1 \cdot j_2 \\ i_1 \cdot k_2 & j_1 \cdot k_2 & k_1 \cdot k_2 \end{bmatrix} \begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} \quad \text{------ (4)}$$

$${}^2P = {}^2_1R \cdot {}^1P$$ < Pt 'p' in frame {1} is transformed to frame {2} >

$${}^2_1R = \begin{bmatrix} i_1 \cdot i_2 & j_1 \cdot i_2 & k_1 \cdot i_2 \\ i_1 \cdot j_2 & j_1 \cdot j_2 & k_1 \cdot j_2 \\ i_1 \cdot k_2 & j_1 \cdot k_2 & k_1 \cdot k_2 \end{bmatrix}$$

Mapping

Mapping between Rotated Frame

$${}^{1}p = {}^{1}_{2}R \cdot {}^{2}p$$

$${}^{2}p = \left[{}^{1}_{2}R\right]^{-1} \cdot {}^{1}p$$

$${}^{2}p = {}^{2}_{1}R \cdot {}^{1}p$$

$${}^{2}p = \left[{}^{1}_{2}R\right]^{T} \cdot {}^{1}p$$

$${}^{2}_{1}R = \left[{}^{1}_{2}R\right]^{-1} = \left[{}^{1}_{2}R\right]^{T}$$

$${}^{1}_{2}R = \begin{bmatrix} i_2 \cdot i_1 & j_2 \cdot i_1 & k_2 \cdot i_1 \\ i_2 \cdot j_1 & j_2 \cdot j_1 & k_2 \cdot j_1 \\ i_2 \cdot k_1 & j_2 \cdot k_1 & k_2 \cdot k_1 \end{bmatrix}$$

$${}^{2}_{1}R = \begin{bmatrix} i_1 \cdot i_2 & j_1 \cdot i_2 & k_1 \cdot i_2 \\ i_1 \cdot j_2 & j_1 \cdot j_2 & k_1 \cdot j_2 \\ i_1 \cdot k_2 & j_1 \cdot k_2 & k_1 \cdot k_2 \end{bmatrix}$$

$$= \left[{}^{1}_{2}R\right]^{T}$$

Inverse of Rotation Matrix = Transpose of Rotation Matrix

$$R^{-1} = R^{T}$$

$$\therefore RR^{T} = I = 3 \times 3 \text{ identity matrix.}$$

Properties of Rotation Matrix

1) The inverse of a rotation matrix is the transpose of it

$R^{-1} = R^{T}$ ← A matrix with this cond^n is called ORTHOGONAL

$RR^{-1} = I$

$\therefore RR^{T} = I =$ Identity matrix (3x3 unit matrix)

[Orthogonality of 'R_3' comes from the fact that it maps an orthogonal co-ordinate frame to another orthogonal co-ordinate frame] : 'orthogonality property.'

2) The determinant of the rotation matrix is unity $\left[\begin{array}{l} +1 \text{ for RH Co-ord's} \\ -1 \text{ for LH } \quad " \end{array}\right.$

i.e.,
$$\begin{vmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{vmatrix} = 1$$

3) The dot product of two different row is zero
Similarly the dot product of two different column is zero

4) The dot product of any row with itself is unity
Similarly the dot product of any column with itself is unity.

5) The rotation matrices generally donot commutative i.e. sequence is important.

$$R_x(\theta) . R_z(\phi) \neq R_z(\phi) R_x(\theta).$$

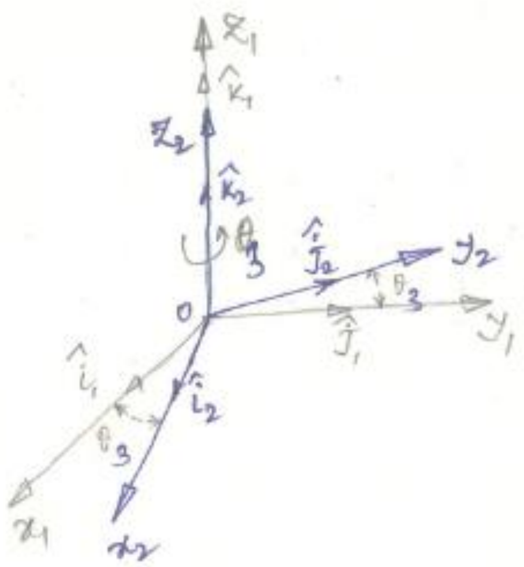
X

Principle Axis Rotation

Fundamental Rotation Matrix

To determine the orientation of frame {2}, which is rotated about the one of the three principal axes of frame {1}. Consider the rotation of frame {2} w.r.t. frame {1} by angle 'θ_3' about the 'z' axis of frame {1}. The corresponding rotation matrix ($^1R_2$) known as the Fundamental Rotation Matrix

$R_z(\theta_3)$



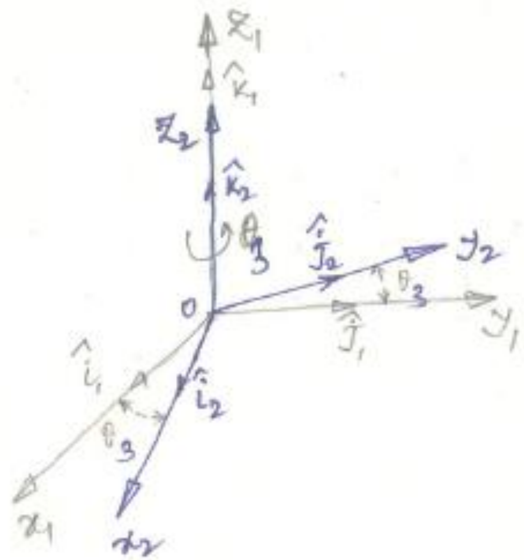
$$\begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} = \begin{array}{c c} & \begin{array}{ccc} x_2 & y_2 & z_2 \end{array} \\ \begin{array}{c} x_1 \\ y_1 \\ z_1 \end{array} & \begin{bmatrix} \hat{i}_2 \cdot \hat{i}_1 & \hat{j}_2 \cdot \hat{i}_1 & \hat{k}_2 \cdot \hat{i}_1 \\ \hat{i}_2 \cdot \hat{j}_1 & \hat{j}_2 \cdot \hat{j}_1 & \hat{k}_2 \cdot \hat{j}_1 \\ \hat{i}_2 \cdot \hat{k}_1 & \hat{j}_2 \cdot \hat{k}_1 & \hat{k}_2 \cdot \hat{k}_1 \end{bmatrix} \end{array} \begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix}$$

Fundamental Rotation Matrix

Rotation about Z

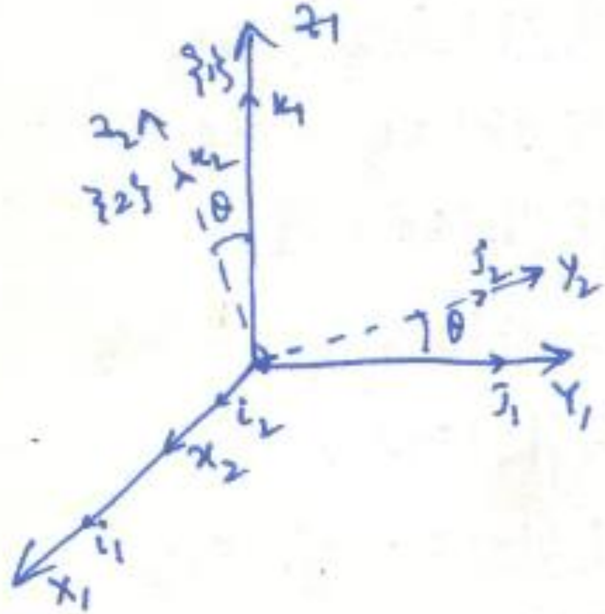
$$\begin{array}{c} \\ \begin{bmatrix} {}^1p_x \\ {}^1p_y \\ {}^1p_z \end{bmatrix} \end{array} = \begin{array}{cc} & \begin{array}{ccc} x_2 & y_2 & z_2 \end{array} \\ \begin{array}{c} x_1 \\ y_1 \\ z_1 \end{array} & \begin{bmatrix} \hat{i}_2\cdot\hat{i}_1 & \hat{j}_2\cdot\hat{i}_1 & \hat{k}_2\cdot\hat{i}_1 \\ \hat{i}_2\cdot\hat{j}_1 & \hat{j}_2\cdot\hat{j}_1 & \hat{k}_2\cdot\hat{j}_1 \\ \hat{i}_2\cdot\hat{k}_1 & \hat{j}_2\cdot\hat{k}_1 & \hat{k}_2\cdot\hat{k}_1 \end{bmatrix} \end{array} \begin{array}{c} \\ \begin{bmatrix} {}^2p_x \\ {}^2p_y \\ {}^2p_z \end{bmatrix} \end{array}$$

$$\hat{i}_2\cdot\hat{i}_1 = |\hat{i}_2||\hat{i}_1|.\cos\theta_3 = \cos\theta_3$$
$$\hat{j}_2\cdot\hat{j}_1 = |\hat{j}_2||\hat{j}_1|\cos\theta_3 = \cos\theta_3$$
$$\hat{k}_2\cdot\hat{k}_1 = |\hat{k}_2||\hat{k}_1|\cos 0^\circ = 1$$
$$\hat{j}_2\cdot\hat{i}_1 = |\hat{j}_2||\hat{i}_1|\cos(90+\theta_3) = -\sin\theta_3$$
$$\hat{k}_2\cdot\hat{i}_1 = |\hat{k}_2||\hat{i}_1|\cos 90^\circ = 0$$
$$\hat{i}_2\cdot\hat{j}_1 = |\hat{i}_2||\hat{j}_1|\cos(90-\theta_3) = \sin\theta_3$$
$$\hat{k}_2\cdot\hat{j}_1 = |\hat{k}_2||\hat{j}_1|\cos 90^\circ = 0$$
$$\hat{i}_2\cdot\hat{k}_1 = |\hat{i}_2||\hat{k}_1|\cos 90^\circ = 0$$
$$\hat{j}_2\cdot\hat{k}_1 = |\hat{j}_2||\hat{k}_1|\cos 90^\circ = 0$$



$$R_z(\theta_3) = \begin{bmatrix} \cos\theta_3 & \cos(90+\theta_3) & \cos 90^\circ \\ \cos(90^\circ-\theta_3) & \cos\theta_3 & \cos 90^\circ \\ \cos 90^\circ & \cos 90^\circ & \cos 0^\circ \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 \\ \sin\theta_3 & \cos\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} C\theta_3 & -S\theta_3 & 0 \\ S\theta_3 & C\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$i_2 \cdot i_1 = |i_2||i_1| \cos 0 = 1$$

$$j_2 \cdot i_1 = 0$$

$$k_2 \cdot i_1 = 0$$

$$i_2 \cdot j_1 = 0$$

$$j_2 \cdot j_1 = C\theta$$

$$k_2 \cdot j_1 = -S\theta$$

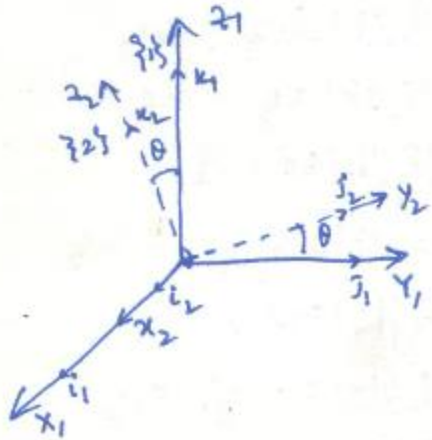
$$i_2 \cdot k_1 = 0$$

$$j_2 \cdot k_1 = S\theta$$

$$k_2 \cdot k_1 = C\theta$$

Fundamental Rotation Matrix

Rotation about X



$i_2 \cdot i_1 = |i_2||i_1| \cos 0 = 1$

$j_2 \cdot i_1 = 0$

$k_2 \cdot i_1 = 0$

$i_2 \cdot j_1 = 0$

$j_2 \cdot j_1 = C\theta$

$k_2 \cdot j_1 = -S\theta$

$i_2 \cdot k_1 = 0$

$j_2 \cdot k_1 = S\theta$

$k_2 \cdot k_1 = C\theta$

$$R_x(\theta_1) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\theta_1 & -S\theta_1 \\ 0 & S\theta_1 & C\theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C_1 & -S_1 \\ 0 & S_1 & C_1 \end{bmatrix}$$

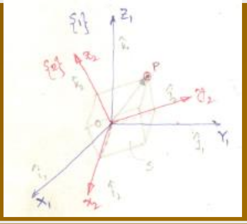
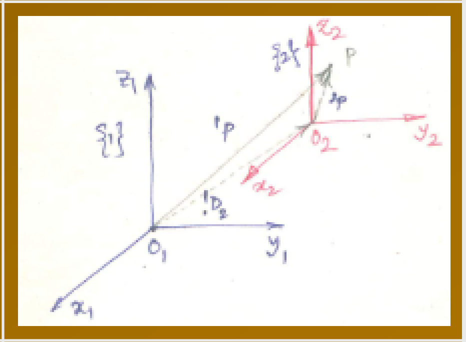
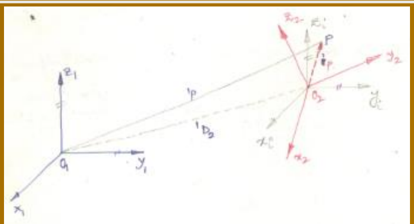
Rotation about Y

$$R_y(\theta_2) = \begin{bmatrix} C\theta_2 & 0 & S\theta_2 \\ 0 & 1 & 0 \\ -S\theta_2 & 0 & C\theta_2 \end{bmatrix} = \begin{bmatrix} C_2 & 0 & S_2 \\ 0 & 1 & 0 \\ -S_2 & 0 & C_2 \end{bmatrix}$$

Mapping

Mapping refers to changing the description of a POINT (or a vector) from one coordinate system (i.e. frame) to another coordinate system (i.e. frame).

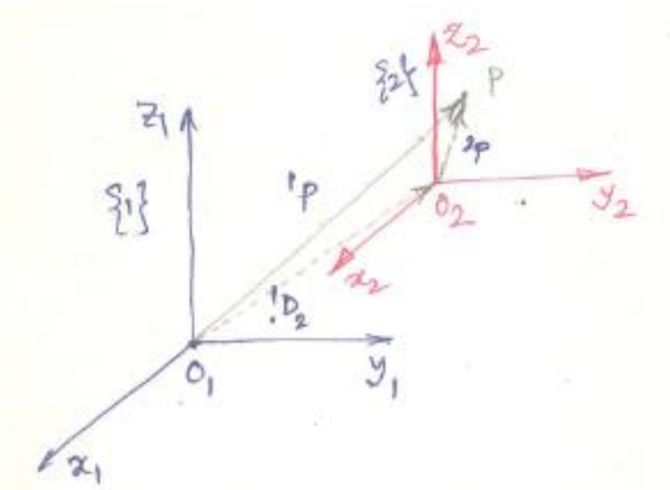
Second coordinate frame has THREE possibilities in relation to the first coordinate frame

(i)	Second coordinate frame is rotated w.r.to the first but origin of both the frames is same	**Mapping between Rotated frame**	
(ii)	Second coordinate frame is moved away from first, the axes of both frames remains parallel. This is a translation of the origin of the second frame from the first frame.	**Mapping between Translated frame**	
(iii)	Second coordinate frame is rotated as well as translated	**Mapping between Rotated & Translated frame**	

Mapping

Mapping between Translated frame

Second coordinate frame is moved away from first, the axes of both frames remains parallel. This is a translation of the origin of the second frame from the first frame.



Consider two frames, frame{1} & frame{2} with origins O_1 & O_2, Such that the axes of frame {1} are parallel to axes of frame {2}.

A pt. 'P' in Space can be expressed as

Vector $\overrightarrow{O_2P}$ $(= {}^2p)$ w.r.to frame {2}

Vector $\overrightarrow{O_1P}$ $(= {}^1p)$ w.r.to frame {1}.

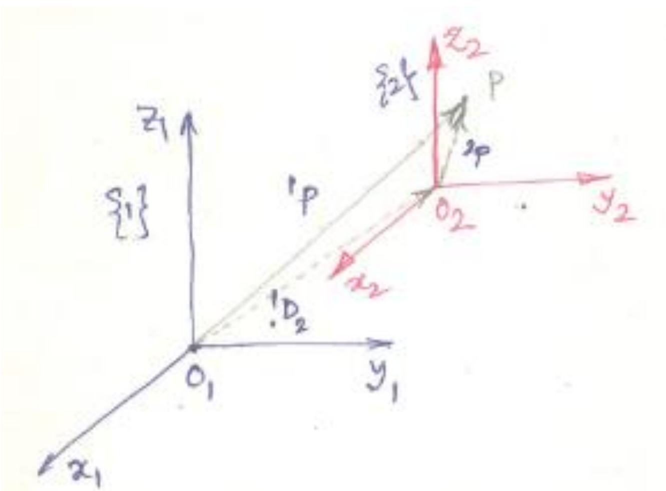
$$\overrightarrow{O_1P} = \overrightarrow{O_2P} + \overrightarrow{O_1O_2}$$

$$^1P = {}^2P + {}^1D_2$$

Mapping

Mapping between Translated frame

Second coordinate frame is moved away from first, the axes of both frames remains parallel. This is a translation of the origin of the second frame from the first frame.



$$\overrightarrow{O_1P} = \overrightarrow{O_2P} + \overrightarrow{O_1O_2}$$

$$^1P = {}^2P + {}^1D_2$$

$$^1p_x = {}^2p_x + d_x = 1\cdot{}^2p_x + 0\cdot{}^2p_y + 0\cdot{}^2p_z + d_x$$

$$^1p_y = {}^2p_y + d_y = 0\cdot{}^2p_x + 1\cdot{}^2p_y + 0\cdot{}^2p_z + d_y$$

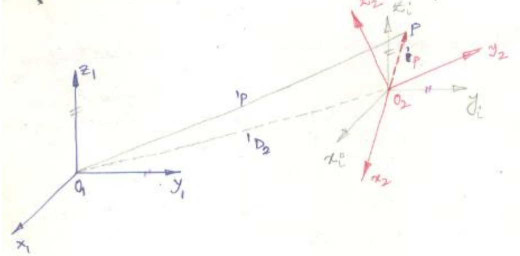
$$^1p_z = {}^2p_z + d_z = 0\cdot{}^2p_x + 0\cdot{}^2p_y + 1\cdot{}^2p_z + d_z$$

$$^1p = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ^2p_x \\ ^2p_y \\ ^2p_z \\ 1 \end{bmatrix}$$

$$^1D_2 = \begin{bmatrix} d_x & d_y & d_z & 1 \end{bmatrix}^T = \begin{bmatrix} d_x \\ d_y \\ d_z \\ 1 \end{bmatrix}$$

(1)

Mapping



$$^1p = {}^1R_2\,{}^2p + {}^1D_2$$

$$^1p = {}^1T_2\,{}^2p$$

Homogeneous Transformation Matrix

Here, $^1T_2$ – is 4×4 matrix referred to as the Homogeneous Transformation or Homogeneous Transform.

– It describes both the position & orientation of fra w.r.t. to frame {1}

$$^1T_2 = \begin{array}{c} \begin{array}{cc} \qquad {}^1R_2 \qquad\qquad & \qquad {}^1D_2 \end{array} \\ \left[\begin{array}{ccc|c} i_2 \cdot i_1 & j_2 \cdot i_1 & k_2 \cdot i_1 & d_x \\ i_2 \cdot j_1 & j_2 \cdot j_1 & k_2 \cdot j_1 & d_y \\ i_2 \cdot k_1 & j_2 \cdot k_1 & k_2 \cdot k_1 & d_z \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \end{array}$$

Scale factor

$$= \left[\begin{array}{c|c} \text{Rotation Matrix } (3\times3) & \text{Translation Matrix } (3\times1) \\ \hline \text{Perspective transformation matrix } (1\times3) & \text{Scale factor } (1\times1) \end{array}\right] = \left[\begin{array}{ccc|c} & {}^1R_2 & & {}^1D_2 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

Basic Homogeneous Transformation

Basic Homogeneous Rotational transformation Matrix

or

Basic Homogeneous Rotation Matrices

or

Basic Homogeneous Transformation for Rotation

$$Rot(x, \theta_1) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_1 & -s_1 & 0 \\ 0 & s_1 & c_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\equiv R_x(\theta_1)$

$\equiv R_{x,\theta_1}$

$$Rot(y, \theta_2) = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ 0 & 1 & 0 & 0 \\ -s_2 & 0 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Rot(z, \theta_3) = \begin{bmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Basic Homogeneous Translational transformation Matrix

or

Basic Homogeneous Translation Matrices

or

Basic Homogeneous Transformation for Translation

$$Trans(x, d_x) = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Trans(y, d_y) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Trans(z, d_z) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotational matrix or Composite

Basic rule for finding Composite or Resultant Homogeneous transformation matrix

Rule 1: If the rotation (or transformation) of a frame is about axis of fixed reference frame then the overall/resultant homogeneous transformation matrix is computing by PRE-MULTIPLICATION of the basic homogeneous transformation matrix.

Case(I) (i) $R = R_{Y,90^\circ}\, R_{Z,90^\circ}$ (ii) $R = R_{Z,90^\circ} \cdot R_{Y,90^\circ}$

Case(I) (i) $R = R_{Y,90^\circ}\, R_{Z,90^\circ}$ (ii) $R = R_{Z,90^\circ} \cdot R_{Y,90^\circ}$

$T = Rot(Y,90^\circ)\, Rot(Z,90^\circ)$ $T = Rot(Z,90)\, Rot(Y,90^\circ)$

Rule 2: If rotation (or transformation) of a frame is about its own axis then the overall/resultant homogeneous transformation matrix is computed by POST-MULTIPLICATION of the basic homogeneous transformation matrix

(ii)

$R_Y R_Z$

Case(II) (i) $R = R_{Z,90^\circ} \cdot R_{Y,90^\circ}$ (ii) $R = R_{Y,90^\circ} \cdot R_{Z,90^\circ}$

Composite Homogeneous Transformation Matrix

Composite Transformation :

fig : Composite Transformation of Three frames

– Three frames, with each frame rotated & translated from its preceding frame.

– It is proposed to find the transform, which relates ^3p in frame {3} to ^1p in frame {1}.

– The transformation matrix $^2T_3$ can be used to map ^3p, the pt. 'p' in frame {3} to frame {2}

$$^2p = {}^2T_3 \cdot {}^3p$$

Similarly

$$^1p = {}^1T_2 \cdot {}^2p$$

$$\therefore \; {}^1p = {}^1T_2 \left({}^2T_3 \cdot {}^3p \right)$$

$$\text{or} \; {}^1p = {}^1T_2 \cdot {}^2T_3 \cdot {}^3p$$

Again

$$^1p = {}^1T_3 \cdot {}^3p$$

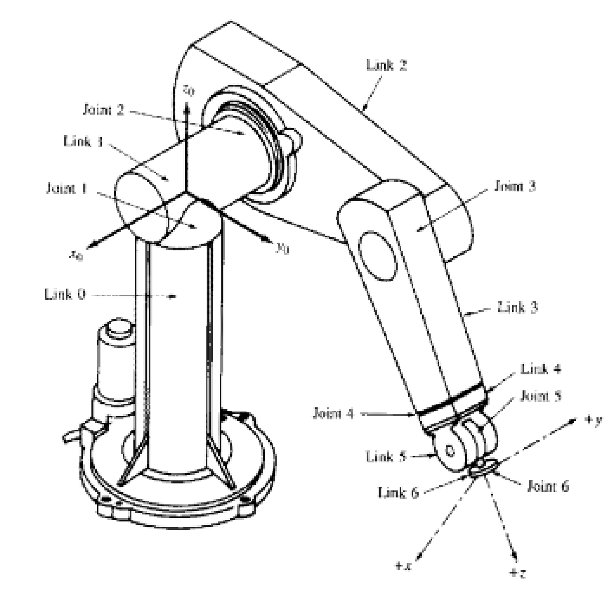
∴ The overall transformation betw. frame {3} & frame {1} is obtained

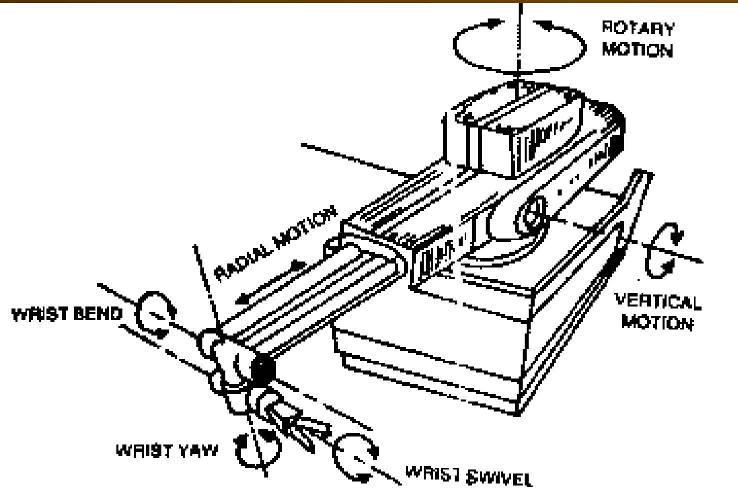
$$^1T_3 = {}^1T_2 \, {}^2T_3$$

It is easily seen that the transformation from frame {1} to frame {i} -

$$^1T_i = {}^1T_2 \, {}^2T_3 \ldots \, {}^{i-1}T_i$$

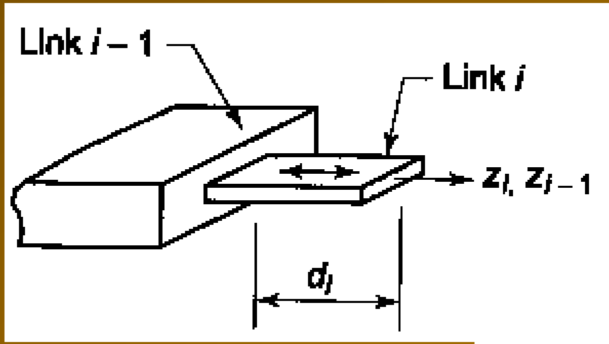
Commonly used Industrial Robotic Manipulator

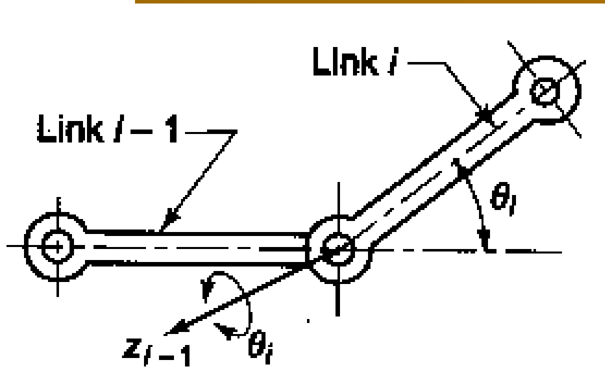




Commonly Joints used in Robotic Mnaipulator

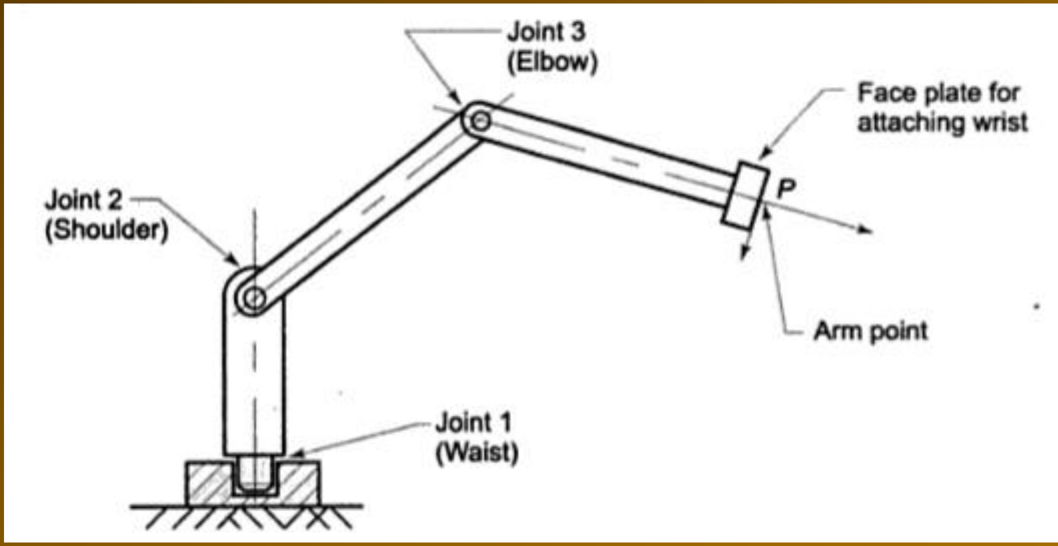
- Linear joint (P)
- Rotary joints (R)





Numbering Convention of Link, Joint & Frame

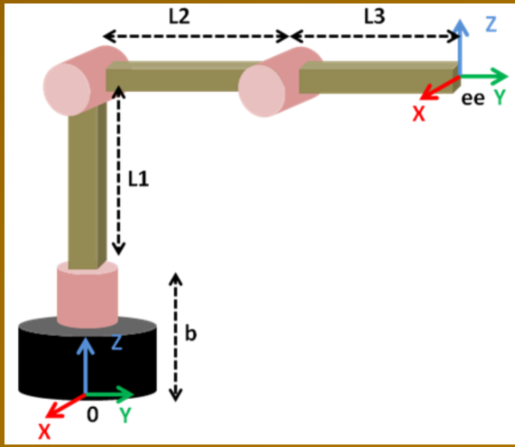
An 'n' DOF robotic arm consists of (n+1) links (including fixed link i.e, base or link 0) connected by 'n' joints of 1 DOF each



Link number: 0 to n

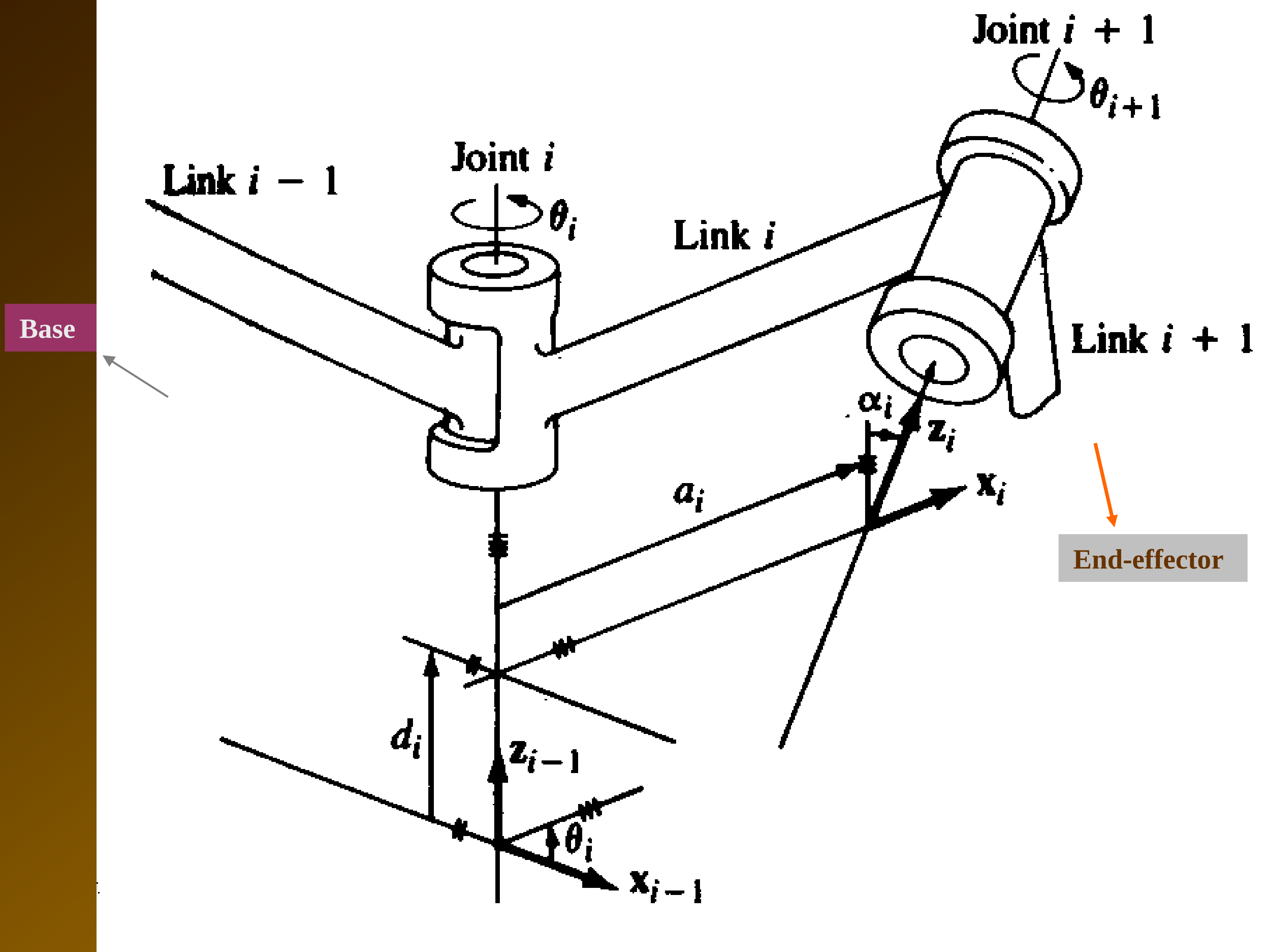
Joint number: 1 to n

Frame number or Axis no. 0 to n



A frame {i} is rigidly attached to distal joint [i.e., joint (i+1)] of link 'i' & it moves with link 'i':

Denavit-Hartenberg (DH) Notation



Denavit-Hartenberg (DH) Notation

A frame {i} is assigned to distal joint [i.e., joint (i+1)] of link 'i' as follows:

1. The z_i axis is aligned with axis of joint (i+1) [i.e., $(i+1)^{th}$ joint axis].

   The positive direction of the z_i axis is arbitrary.

2. The x_i axis is defined along the common normal between the z_{i-1} and z_i axis & pointing from the z_{i-1} to the z_i axis.

   [Common normal has the shortest distance between the two skew lines]

   The origin of frame {i} rigidly attached to the link 'I' is located at the intersection of z_i & x_i axis

Three more possible cases:

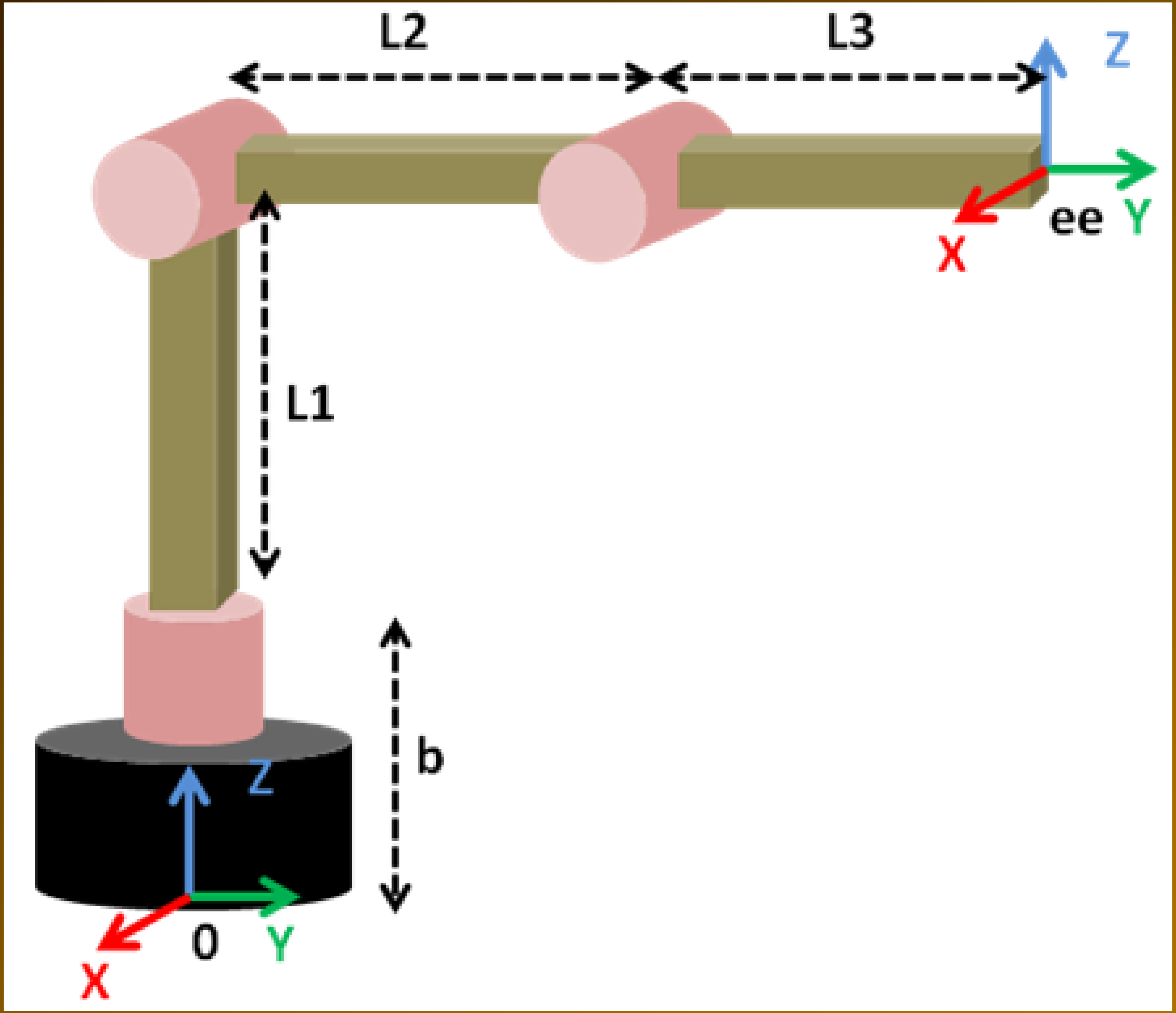
Case (i) when the two consecutive z-axes (z_{i-1} and z_i) are intersecting, then x_i –axis will be perpendicular to the plane formed by z_{i-1} and z_i axes in the direction of $z_{i-1} \times z_i$

The origin of frame {i} is located at the point of their intersection

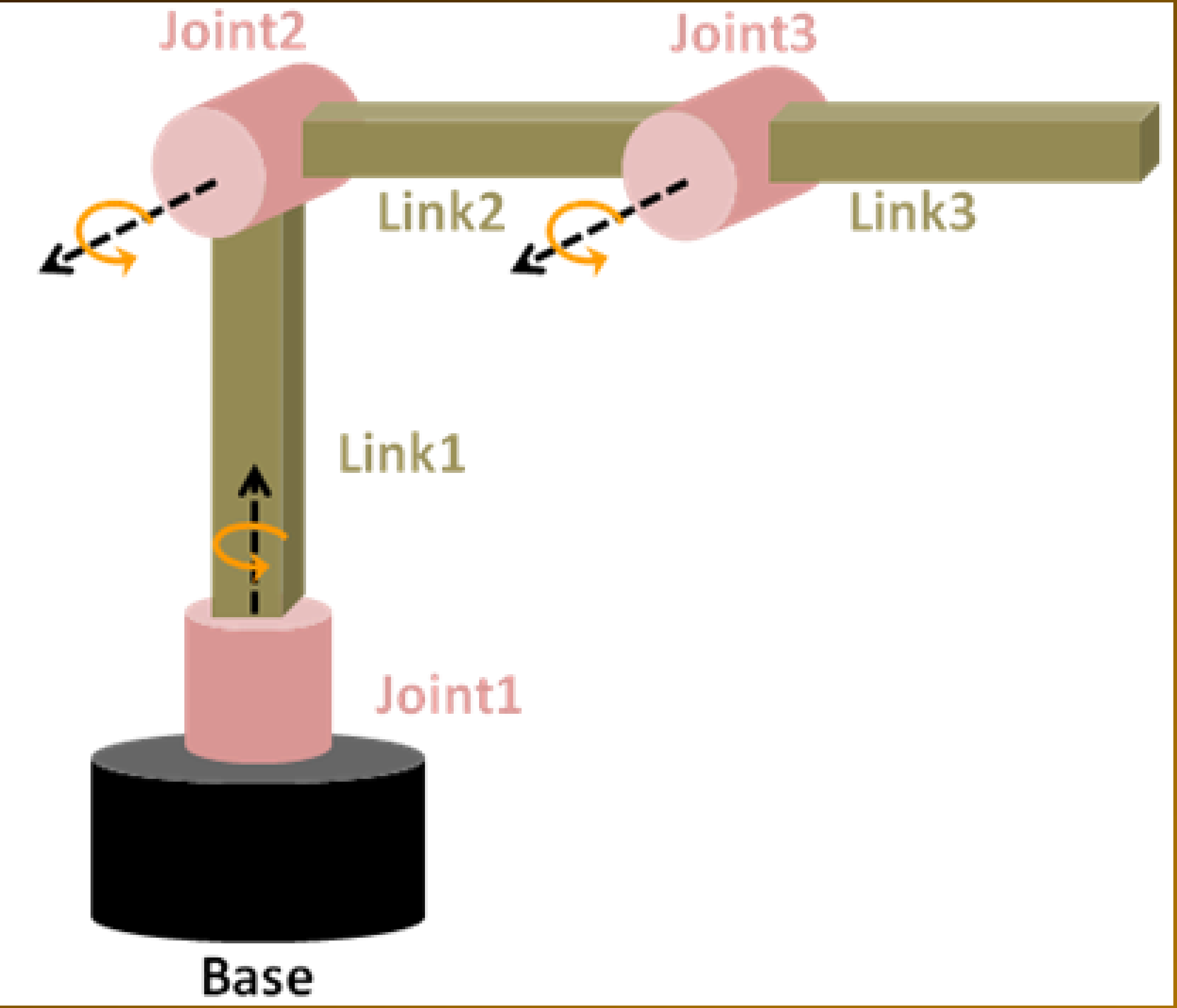
Case (ii) when the two consecutive z-axes (z_{i-1} and z_i) are parallel, then there are an infinite number of common normal. You may consider any one. But prefer to choose that common normal which is passing through origin of frame {i-1}

Case (iii) when the two consecutive z-axes (z_{i-1} and z_i) are collinear, then x_i –axis is chosen parallel to previous x axis i.e, x_{i-1}

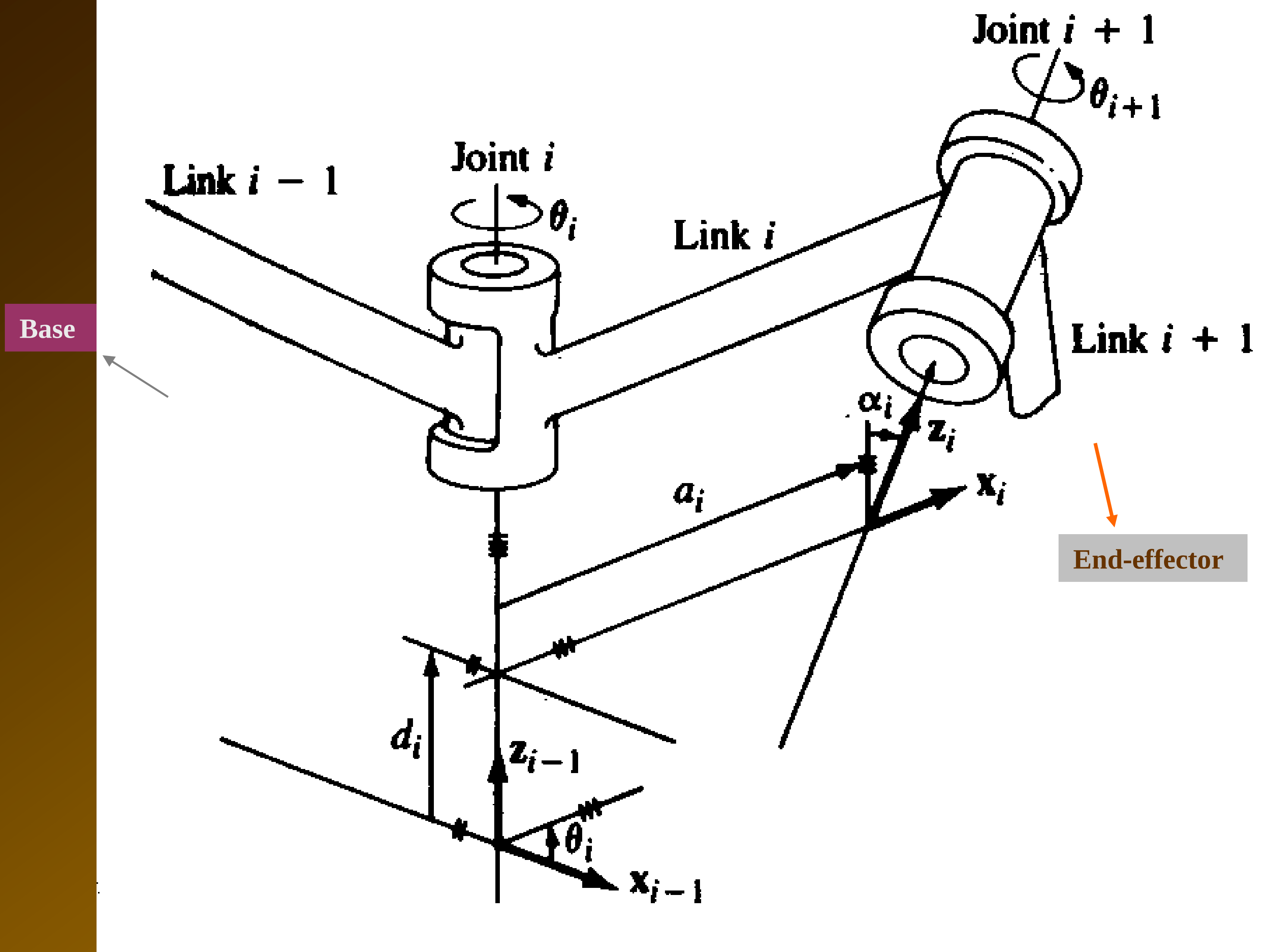
3. Finally, the y_i –axis is determined by the RH rule (i.e., $y_i = z_i \times x_i$)



L2
L3
Z
ee
Y
X
L1
b
Z
0
Y
X


Joint2
Joint3
Link2
Link3
Link1
Joint1
Base

Denavit-Hartenberg (DH) Notation



D-H parameters
or
Joint-link parameters

- Link parameters
  - *Link length (a_i)*
  - *Link twist (α_i)*
- Joint parameters
  - Joint distance (d_i)
  - Joint angle (θ_i)

Link parameters

Determine the structure of the link

Joint parameters

Determine the relative position of neighboring links.

D-H parameters or Joint-link parameters

Link parameters

Link length (a_i) of Link 'i'

- Shortest distance (along common normal) between two motion axes of a link 'i'

- Distance between z_{i-1} & z_i axes along the x_i – axis.

NB: Link length is not the physical length of the rigid link rather it is perpendicular distance between two axes of the binary link

Always POSITIVE

Link twist (α_i) of Link 'i'

- The angle between projection of axis (i-1) and axis i on a plane perpendicular to the common normal.

- Angle between z_{i-1} & z_i axes measured from z_{i-1} about x_i – axis in RH sense.

- or required rotation of the z_{i-1} axis about the x_i – axis to become parallel to the z_i axis

Sign convension: RH rule [CCW (+), CW (-)]

D-H parameters or Joint-link parameters

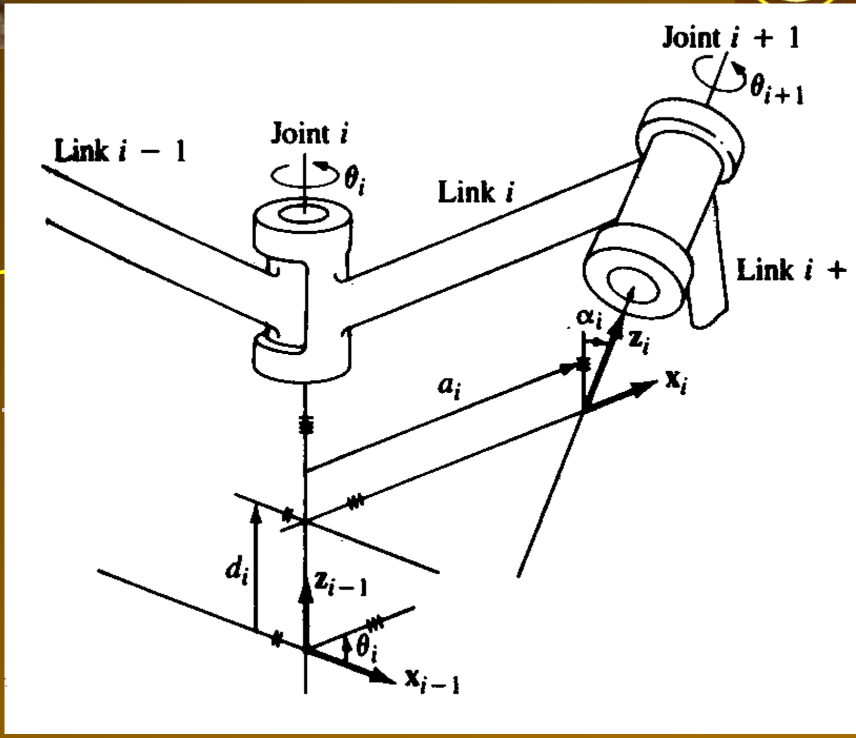
Joint parameters

Joint distance (d_i) of joint 'I'

- The perpendicular distance between two adjacen common normal measured along motion axis of joint 'i'

- The distance between x_{i-1} & x_i axes along the z_{i-1} - axis.

Signed quantity



Joint angle (θ_i) of joint 'I'

- The angle between the two adjacent common normal about axis of joint 'i' in RH sense

- Angle between x_{i-1} & x_i axes measured from x_{i-1} about positive z_{i-1} in RH sense.

- or required rotation of the x_{i-1} axis about the z_{i-1} – axis to become parallel to the x_i axis

Sign convension: RH rule [CCW (+), CW (-)

D-H parameters
or
Joint-link parameters

- **Link parameters**
  - *Link length (a_i)*
  - *Link twist (α_i)*
- **Joint parameters**
  - Joint distance (d_i)
  - Joint angle (θ_i)

Fixed parameters

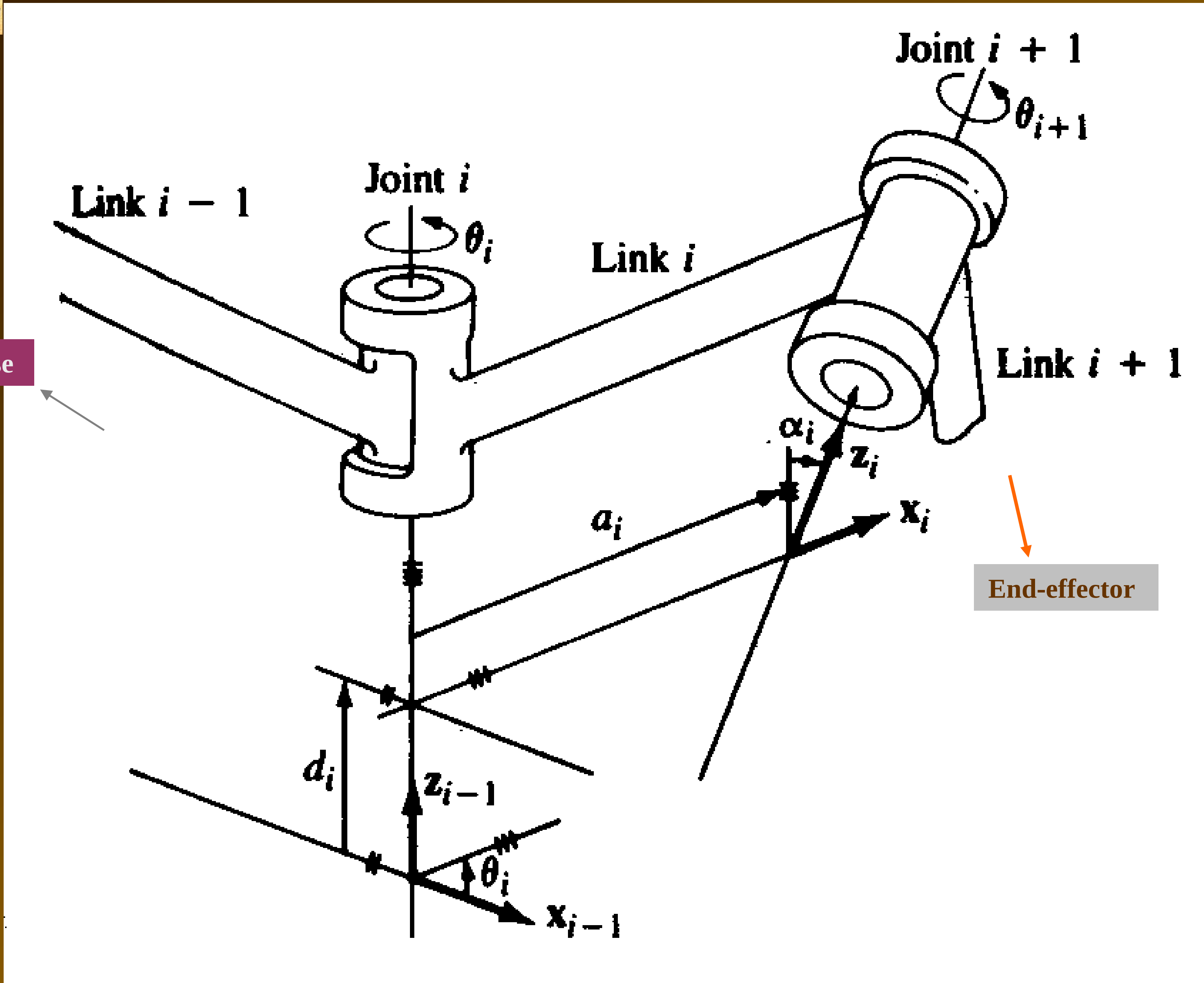
- Both Link parameters

Variable parameters

- Either of two joint parameters

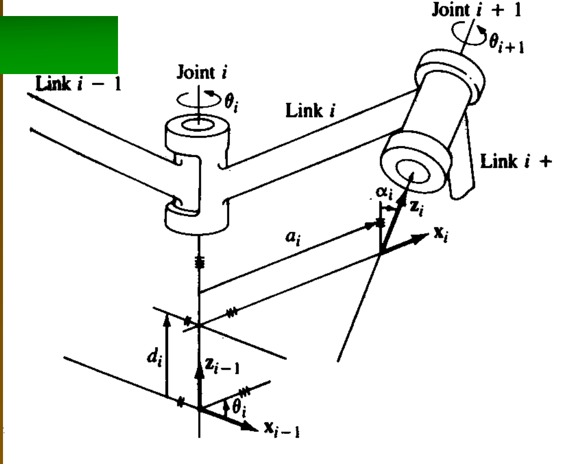
Joint Variable (JV)

- Joint distance (d_i) for prismatic or linear joint (but joint angle=constant)
- Joint angle (θ_i) for rotary joint (but joint distance=constant)


Joint $i+1$
θ_{i+1}
Joint i
θ_i
Link $i-1$
Link i
Link $i+1$
Base
α_i
z_i
a_i
x_i
End-effector
d_i
z_{i-1}
θ_i
x_{i-1}

Kinematic Relationship between adjacent links of a manipulator

The transformation of frame $\{i-1\}$ to frame $\{i\}$ consists of four basic transformations as shown in Fig. 3.9.



A rotation about z_{i-1}-axis by an angle θ_i;

$$\begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & 0 \\ \sin\theta_i & \cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translation along z_{i-1}-axis by distance d_i;

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translation by distance α_i along x_i-axis,

$$\begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation by an angle α_i about x_i-axis.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_i & -\sin\alpha_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Each of these four operations can be expressed by a basic homogeneous rotation-translation matrix and the product of these four basic homogeneous transformation matrices yields a composite homogeneous transformation matrix,

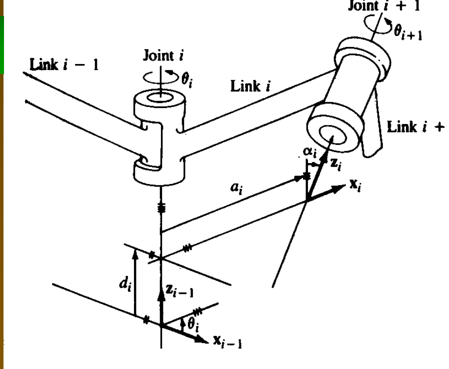
$$^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & 0 \\ \sin\theta_i & \cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_i & -\sin\alpha_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

D-H transformation matrix

link i transformation matrix.

Kinematic Relationship between adjacent links of a manipulator



D-H transformation matrix

link i transformation matrix.

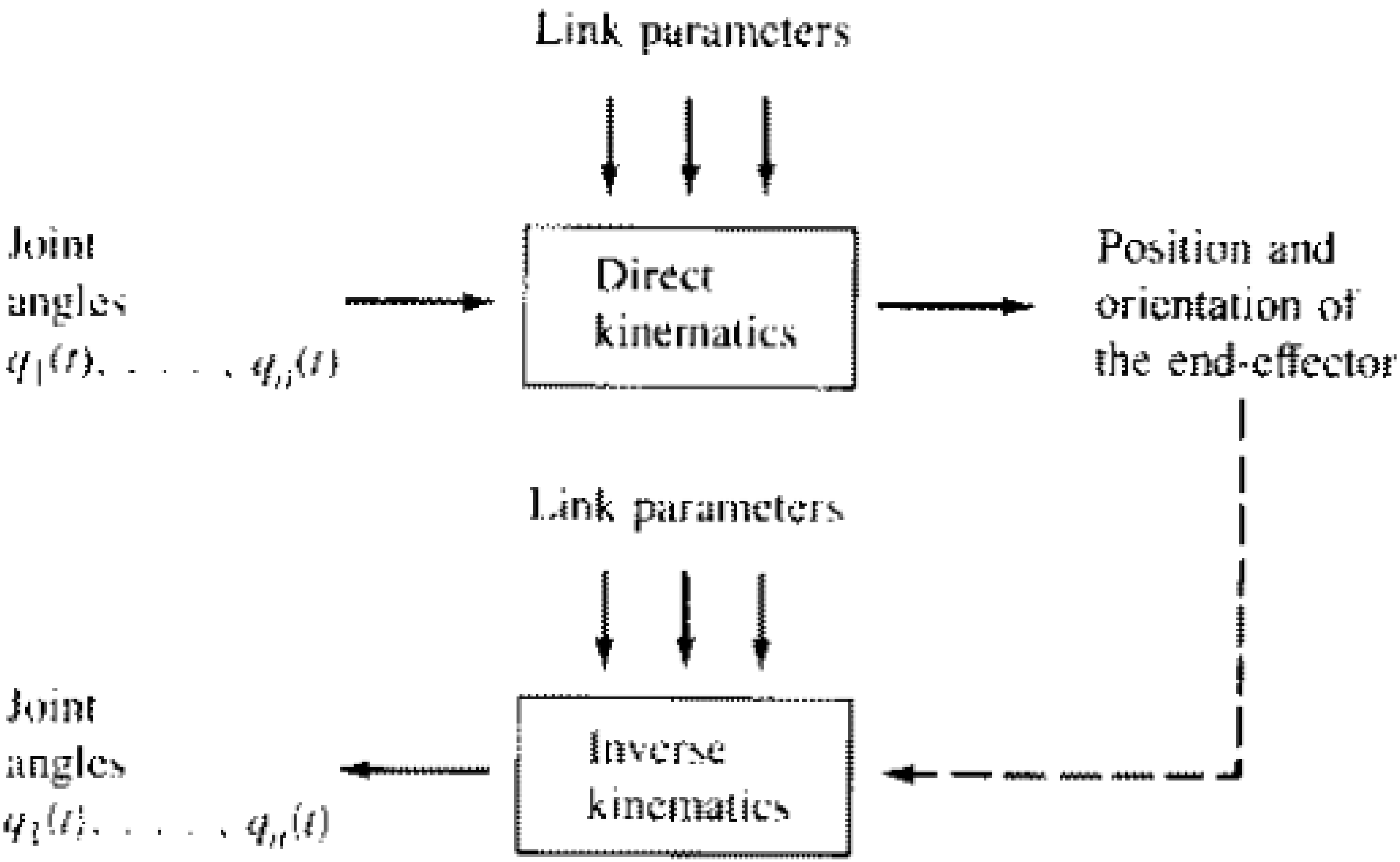
$$
{}^{i-1}T_i = \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

where $C\theta_i = \cos\theta_i$, $S\theta_i = \sin\theta_i$, $C\alpha_i = \cos\alpha_i$, and $S\alpha_i = \sin\alpha_i$.

$$
{}^{i-1}T_i = \left[\begin{array}{ccc:c} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]
$$

The homogeneous transformation matrix ${}^{i-1}T_i$ describes the position and orientation of frame $\{i\}$ relative to frame $\{i\text{–}1\}$ and completely specifies the geometric relationship between these links in terms of four DH-parameters $(\theta_i, d_i\ \alpha_i\ a_i)$. Of these four parameters, only one is a variable for link i, the displacement variable q_i (θ_i or d_i) and other three are constant. The matrix ${}^{i-1}T_i(q_i)$ is known as link i transformation matrix. As shown before, the 3×3 upper left corner submatrix of Eq. (3.5) gives the orientation of coordinate axes of frame $\{i\}$, while the 3×1 upper right corner sub-matrix represents the position of the origin of frame $\{i\}$.

Robot Arm Kinematics



Forward or Direct Kinematics Robotic Arm

Main Steps in Forward Kinematics of Robotic Arm

Step 1 - Choose suitable home position & Draw Kinematic diagram

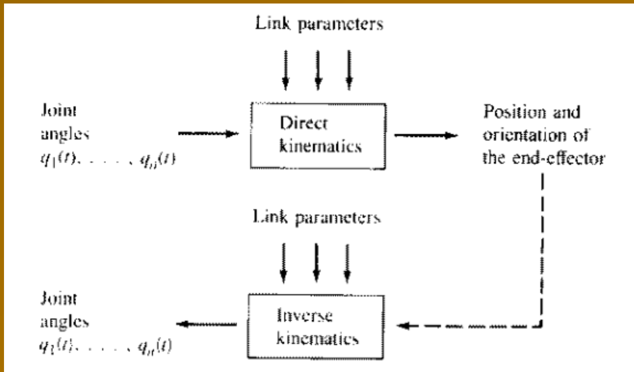
Step 2 - Numbering the links (o to n), numbering the joint (1 to n) & identify the motion axis

Step 3 - Assign RH frame to each link of the robotic manipulator arm starting from base to the end,, based on DH notation

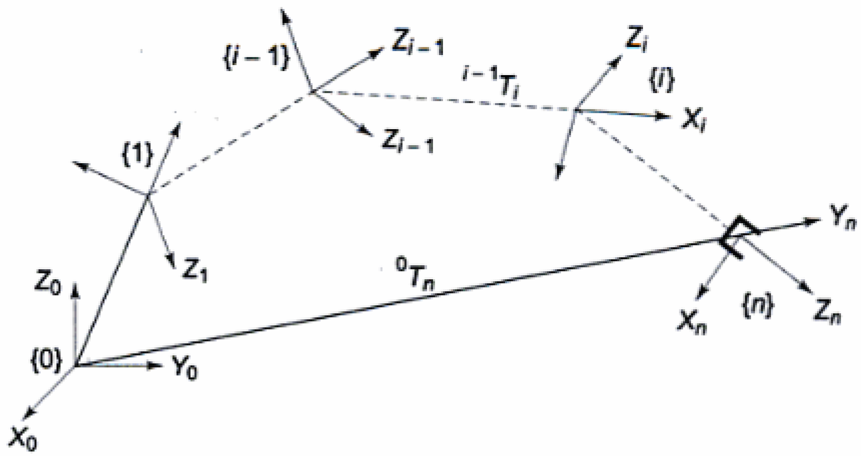
Step 4 - Write or prepare D-H parameters table .

Step 5 - Evaluate individual Link Transformation Matrix or DH transformation matrix of all links (i.e., ${}^{0}T_{1}$, ${}^{1}T_{2}$, ${}^{2}T_{3}$ …… ${}^{n-1}T_{n}$)

Step 6 - Evaluate Composite HTM or Manipulator Arm Transformation Matrix (${}^{0}T_{n}$).



MANIPULATOR TRANSFORMATION MATRIX



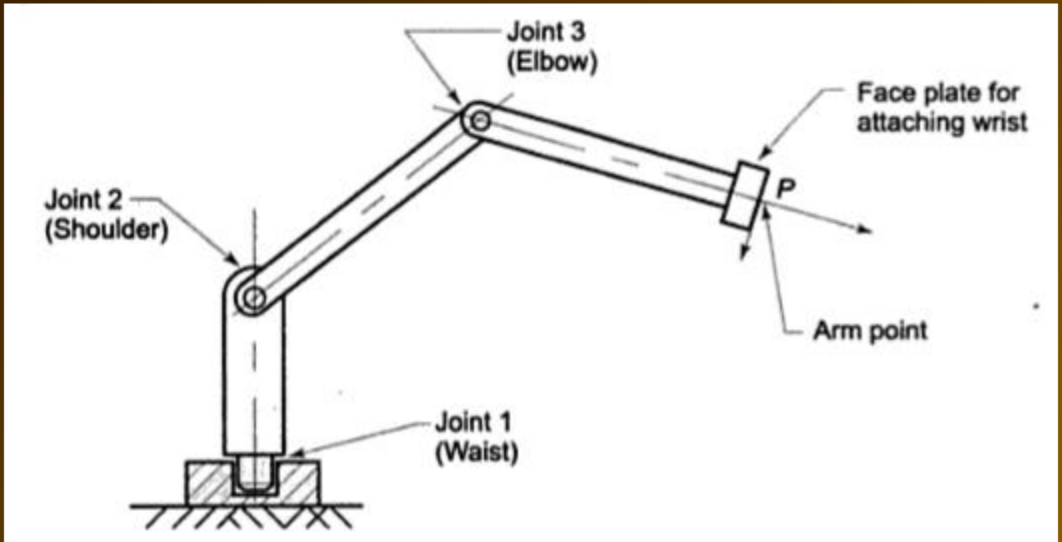
The position and orientation of the tool frame relative to the base frame can be found by considering the n consecutive link transformation matrices relating frames fixed to adjacent links. Thus,

$$ {}^{0}T_{n} = {}^{0}T_{1}(q_1)\ {}^{1}T_{2}(q_2)\ \dots\ {}^{n-1}T_{n}(q_n) \tag{3.6}$$

where ${}^{i-1}T_i(q_i)$ for $i = 1, 2, \dots, n$ is the homogeneous link transformations matrix between frames $\{i–1\}$ and $\{i\}$ and is given by Eq. (3.5).

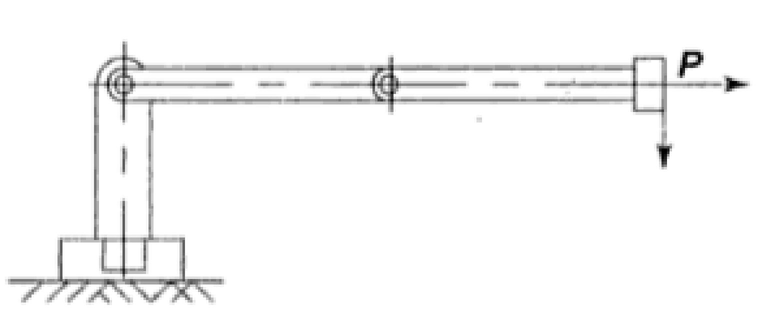
$$ {}^{0}T_{n} = {}^{0}T_{1}\ {}^{1}T_{2}\ \dots\ {}^{n-1}T_{n} $$

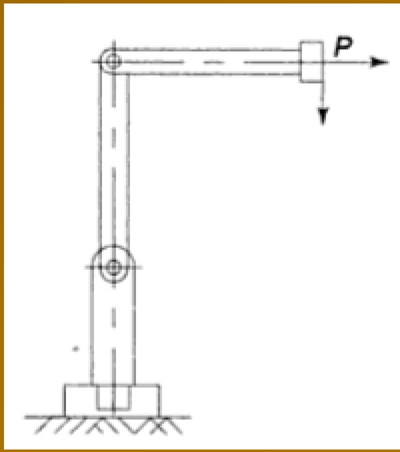
Forward or Direct Kinematics of 3 DOF spatial robotic arm

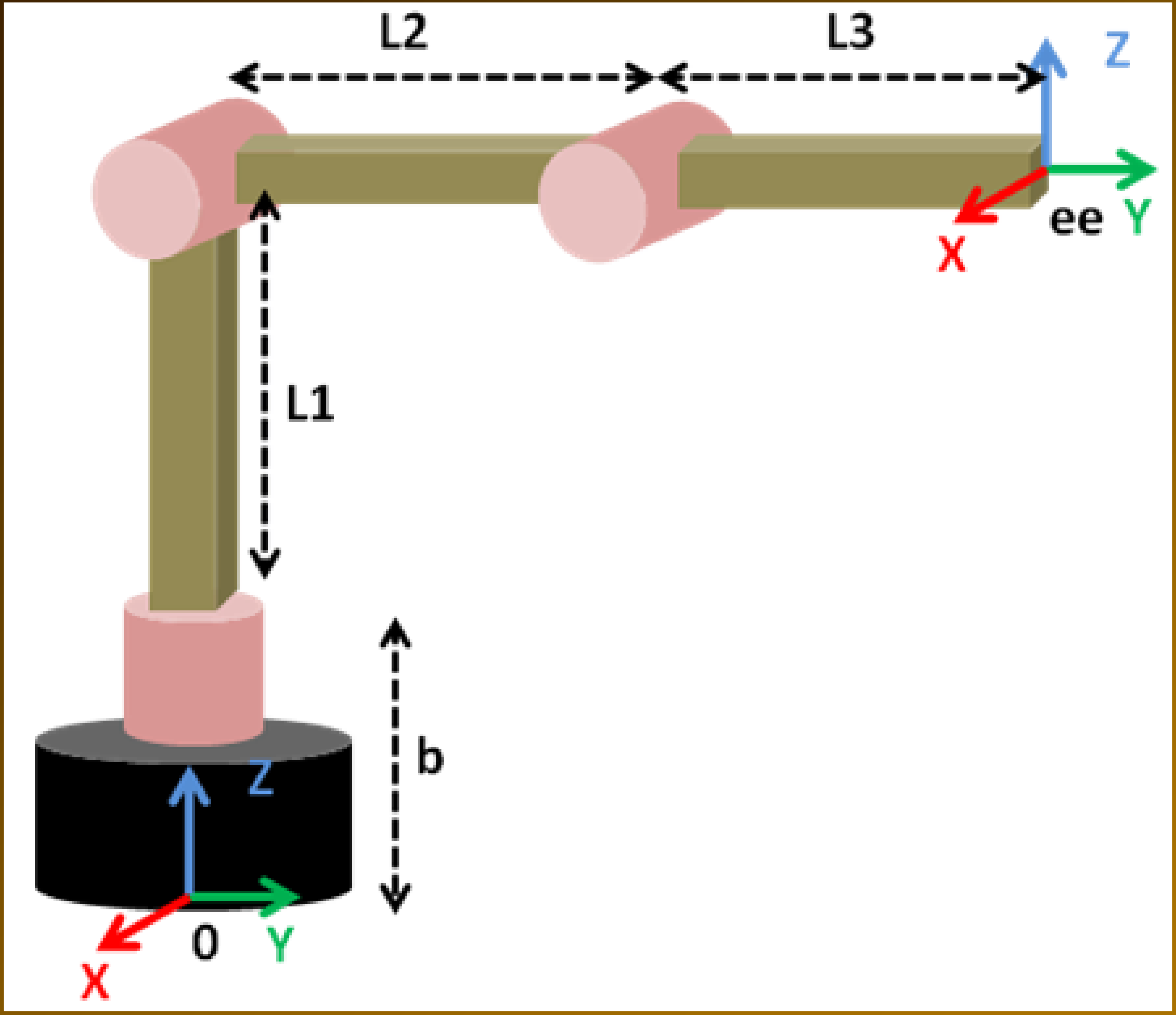


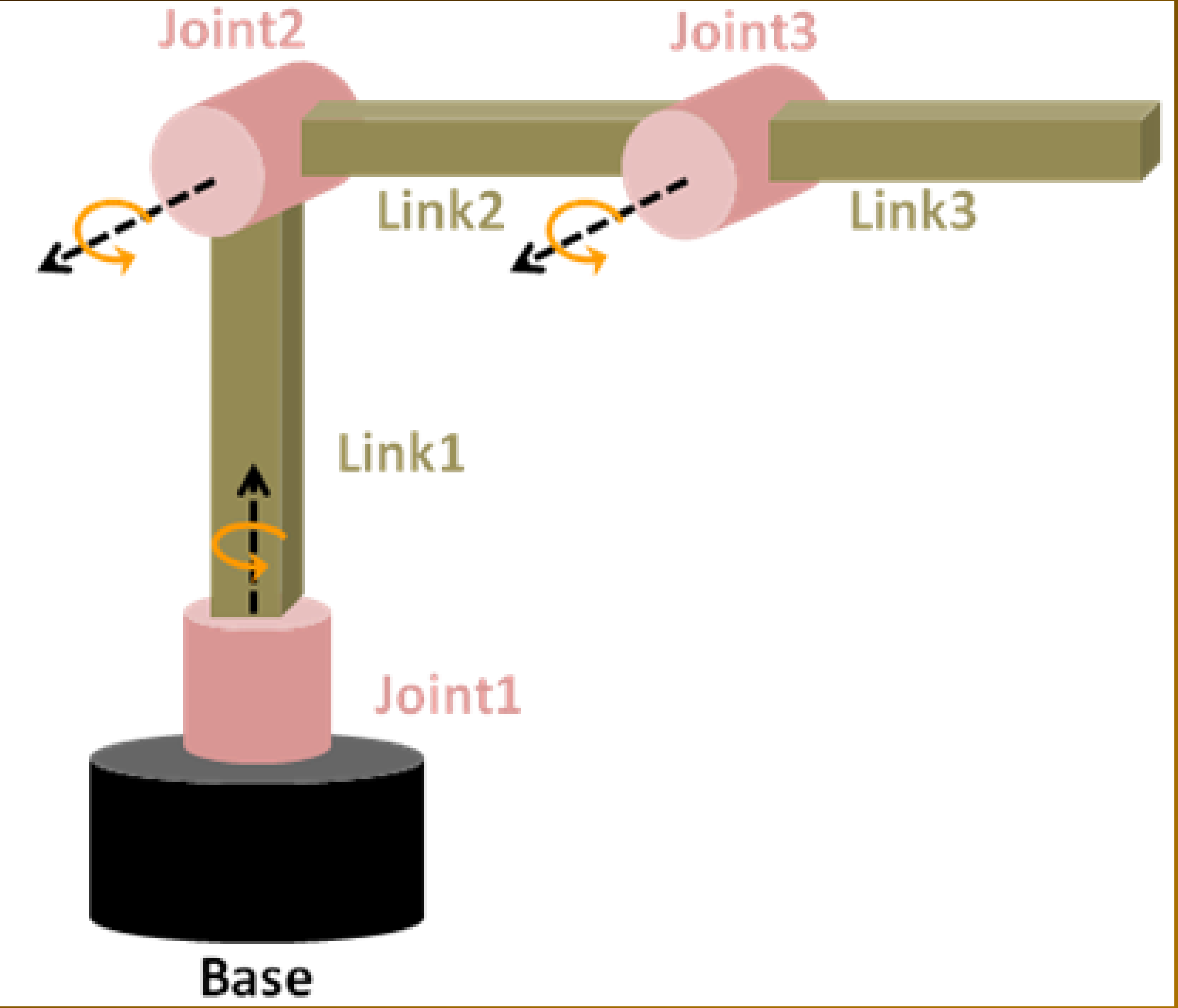
Step 1 - Choose suitable home position & Draw Kinematic diagram

Step 2 - Numbering the links (o to n), numbering the joint (1 to n) & identify the motion axis



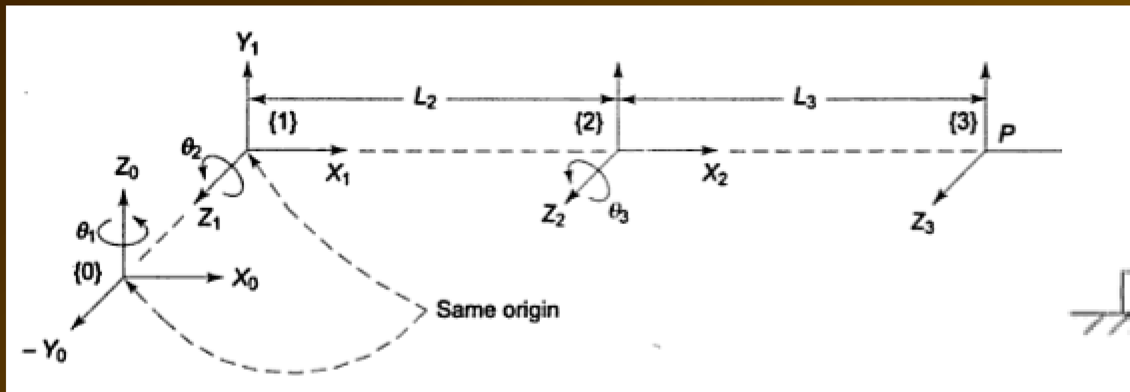


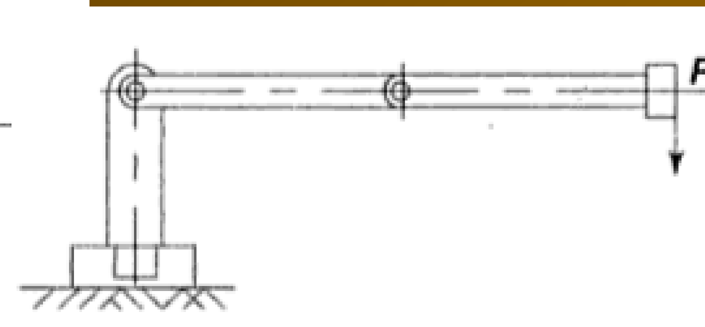

L2
L3
Z
Y
X
ee
Y
L1
b
Z
X
0
Y



Joint2
Joint3
Link2
Link3
Link1
Joint1
Base

Step 3 - Assign RH frame to each link of the robotic manipulator arm starting from base to the end, based on DH notation





Step 4 - Write or prepare D-H parameters table .

Link no., Joint no., Frame no. (i)	Link length (a_i)	Link twist (α_i)	Joint distance (d_i)	Joint angle (θ_i)
1	a_1=0	α_1=90	d_1=95	θ_1 (JV)
2	a_2=100	α_2=0	d_2=0	θ_2 (JV)
3	a_3=75	α_3=0	d_3=0	θ_3 (JV)